Magic squares as orthogonality conditions on a height function

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Abstract

Place the centred entries of an $n \times n$ magic square as a height function z over the centred cell grid. The magic conditions are then the orthogonality of z to a $2n$-dimensional space W of centred line indicators. This viewpoint gives several exact results. (i) A moment hierarchy: orthogonality to x^m, y^m for $m \le k$ is equivalent to equal row and column sums exactly when $k \ge n-1$ for real heights, whereas for permutations of $1, \dots, n^2$ integrality lowers the threshold to at most $n-2$ for every $n \ge 14$. (ii) Balance of the signed mass z: $z \perp W$ forces every separable moment, including Loly's moment-of-inertia invariant, to vanish; for an arbitrary arrangement the centre-of-mass offset is bounded by the line-sum energy with a sharp constant; balance of unit vectors is not implied (exact values for the eight 3×3 squares). (iii) Exact mean and variance of the energy of a uniformly random arrangement, with a limit law. (iv) An exact single-swap formula, giving strict local minimality of every magic square and gaps d^2m/n^4. (v) The exact number 14,639,440 of 4×4 arrangements satisfying four low-order conditions, of which 7,040 are magic.

Keywords: magic squares, orthogonality, permutation statistics, moment of inertia, local search
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1 Introduction

Magic squares are among the oldest objects of recreational mathematics: the 3×3 Lo Shu appears in Chinese sources more than two millennia old, a 4×4 square is engraved in Dürer's *Melencolia I* (1514), and systematic constructions for every order have been known since the seventeenth century [2, 17]. The modern literature treats them from several directions: enumeration and the geometry of the associated polytopes [20, 7, 1, 4, 3], spectral and matrix properties [13, 15, 16, 5], and, in work of Loly and collaborators, physical invariants such as the moment of inertia and the multipole moments [11, 18, 19].

Definition 1.1 (Magic square)**.** An *arrangement* of order n is a bijection $S = (s_{ij})$ from the cells of the $n \times n$ grid onto $\{1, \dots, n^2\}$. It is a *magic square* if every row, every column and both main diagonals sum to the magic constant $M(n) = n(n^2+1)/2$, and *semi-magic* if only the rows and columns do.

This paper reads an arrangement as a function on the grid: the centred entry $z_{ij} = s_{ij} - (n^2+1)/2$ is a *height* over the centred cell $(x_{ij}, y_{ij}) = (j - \frac{n-1}{2},\ \frac{n-1}{2} - i)$ (Definition 2.1). The points $(x_{ij}, y_{ij}, z_{ij}) \in \mathbb{R}^3$ have a convex hull, studied in the companion paper A [14]; here the object is the height function z alone. Every line-sum condition "row k sums to $M(n)$" becomes "z is orthogonal to the centred indicator of row k", i.e. $\mathrm{Cov}(R_k, Z) = 0$ for the empirical covariance over the n^2 cells. A magic square is therefore an arrangement whose height function is orthogonal to a fixed $2n$-dimensional space W of test functions, and the *energy* $E_{\mathrm{full}}(S) = \sum_L \mathrm{Cov}(\mathbf{1}_L, Z)^2$

over the $2n+2$ lines vanishes exactly on magic squares (Lemma 3.8). This is a reformulation of the definition, not a theorem about magic squares (Remark 3.9); its value is that it makes certain questions natural, and the paper is organised around their answers.

Contributions.

1. *The moment hierarchy* (Proposition 3.11, Theorem 3.15). Replacing the line indicators by the polynomial test functions x^m, y^m gives, for each degree k, the conditions $\mathrm{Cov}(X^m, Z) = \mathrm{Cov}(Y^m, Z) = 0$ for $m \le k$. For real-valued height functions these are equivalent to equal row and column sums if and only if $k \ge n-1$ (a Vandermonde fact). For permutations of $1, \dots, n^2$, integrality and the bound on line sums lower the threshold $k^*(n)$ to at most $n-2$ for every $n \ge 14$, whereas $k^*(n) = n-1$ for $3 \le n \le 13$ (computer-verified witnesses, Remark 3.16). The four-term energy E_{low} is the degree-one truncation. Loly's invariance of the moment of inertia [11] is the single condition $\mathrm{Cov}(Z, X^2+Y^2) = 0$, implied by the degree-two conditions but much weaker than them (Corollary 3.13); the same identity gives the null distribution of the moment of inertia of a random arrangement (Corollary 3.14).
2. *Balance of the signed mass* (Section 4). $E_{\mathrm{full}} = 0$ is equivalent to $\sum_c z_c \varphi_c = 0$ for every φ in the line space (Theorem 4.1); this contains Loly's invariant, the vanishing of the centre of mass and of all axial moments, and leaves the mixed moments free. For any arrangement, $|\sum_c z_c(x_c, y_c)|^2 \le N^2 \frac{n(n^2-1)}{12} E_{\mathrm{full}}$ with a sharp constant (Theorem 4.2). The corresponding statement for unit vectors is false: the eight 3×3 magic squares all have a ring imbalance $2(2-\sqrt{2})\sqrt{5}$, and a three-dimensional unit-vector sum never vanishes (Proposition 4.3); a numerical comparison for $n = 4$ is given in Table 2.
3. *Exact statistics of a random arrangement* (Proposition 5.1). $\mathbb{E}[E_{\mathrm{low}}] = (n^2+1)(n^4 - n^2 + 12n - 12)/(72n^2)$ and $\mathbb{E}[E_{\mathrm{full}}] = (n^4-1)/(6n^2)$, with exact variances; $E_{\mathrm{low}}/\mathbb{E}[E_{\mathrm{low}}] \to \mathrm{Exp}(1)$ by Hoeffding's combinatorial central limit theorem. Monte Carlo with 10^8 samples per order agrees for $3 \le n \le 8$ to within 5×10^{-4} relative error.
4. *The single-swap formula* (Theorem 3.18). Exchanging two entries of a magic square that differ by d raises E_{full} by exactly $d^2 m/n^4$, where $m \in \{2, \dots, 6\}$ counts the lines separating the two cells; every magic square is a strict local minimum, with minimum gap between $2/n^4$ and $6/n^4$. Elementary steepest descent and annealing on E_{full} are reported in Section 6.2.
5. *The low-mode class for $n = 4$* (Proposition 5.4). Exactly 14,639,440 of the 16! arrangements satisfy $E_{\mathrm{low}} = 0$, of which 7,040 are magic; the count is a meet-in-the-middle enumeration and agrees with a Gaussian local-limit estimate to within 20%.

Context and scope. The enumerative and polyhedral theory of magic squares is classical. Non-negative integer matrices with all line sums equal to t are the lattice points of the t-th dilate of the Birkhoff polytope and their number is a polynomial in t [20, 7]; the count of magic squares with distinct entries is an Ehrhart quasi-polynomial of an inside-out polytope [4, 3]; the cones of magic squares and cubes have Hilbert-basis descriptions [1]; and arrays with fixed margins have been studied as a sample space by Diaconis and Gangolli [6]. None of this is used or extended here: our sample space is the uniform distribution on permutations, and we count nothing polyhedrally. The orthogonality reading is elementary and implicit in Loly's work: he derived the invariance of the moment of inertia from row and column sums alone [11], and Rogers and Loly studied the multipole expansion of a magic cube [19]. The linear balance statements of Section 4 belong to that circle of ideas; what we add is their formulation as a single orthogonality statement, the sharp quantitative bound, and the negative results for nonlinear (unit-vector) balance. The Vandermonde threshold of the hierarchy is elementary linear algebra. To our knowledge the permutation refinement (Theorem 3.15), the exact moments and limit law for random arrangements, the swap formula and the exact count for $n = 4$ are new; they are small statements about a classical object, and we make no claim that the energy reformulation by itself is a result.

Organisation. Section 2 fixes coordinates and collects the facts used; Section 3 contains the projection lemma, the hierarchy and the swap formula; Section 4 the balance results; Section 5 the computations; Section 6 what the viewpoint does and does not buy, local search, and open questions; the appendices the methods, the variance closed forms and a witness for $k^*(5) = 4$.

2 Background

2.1 Centred coordinates and the height function

Let $S = (s_{ij})$ be an arrangement of order n, with cells indexed by $i, j \in \{0, \dots, n-1\}$ (i the row from the top, j the column from the left), and write $N = n^2$.

Definition 2.1 (Centred coordinates and height function)**.** To the cell (i, j) assign

$$x_{ij} = j - \frac{n-1}{2}, \qquad y_{ij} = \frac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \frac{N+1}{2}. \tag{1}$$

The function $z = (z_{ij})$ on the grid is the *height function* of S.

The horizontal coordinates centre the grid at the origin; the vertical coordinate centres the entries so that the mean entry $(N+1)/2$ sits at height zero. All three have mean zero over the N cells (Lemma 3.1). The N points $(x_{ij}, y_{ij}, z_{ij}) \in \mathbb{R}^3$ have a convex hull, the object of the companion paper A, *Magic Gems: A Polyhedral Framework for Magic Squares* (Mathewson, arXiv:2512.09170) [14], which studies the convex hull of these points, the symmetry classification of the 880 order-four hulls, and the hulls of Siamese squares. Nothing below uses the hull.

2.2 Covariance and Loly's planar moment

For functions φ, f on the cells, $\mathrm{Cov}(\varphi, f)$ denotes the empirical covariance over the N cells, $\mathrm{Cov}(\varphi, f) = \frac{1}{N}\sum_c (\varphi_c - \bar{\varphi})(f_c - \bar{f})$, and X, Y, Z denote the functions (1). Loly's planar moment of inertia [11] places the *value* s_{ij} as a mass at the planar point (x_{ij}, y_{ij}) and computes $\sum s_{ij}(x_{ij}^2 + y_{ij}^2)$. It is an affine function of the single degree-two cross-moment $\mathrm{Cov}(Z, X^2 + Y^2)$ (Corollary 3.13).

2.3 Two tools from permutation statistics

Let π be a uniformly random permutation of $\{1, \dots, N\}$ and $a, b \in \mathbb{R}^N$ fixed vectors. The quantity $T_a = \sum_i a_i b_{\pi(i)}$ is a *linear permutation statistic*. Its mean is $N\bar{a}\bar{b}$ and, for two such statistics with the same b (Wald–Wolfowitz [22], polarised),

$$\mathrm{Cov}(T_a, T_{a'}) = \frac{1}{N-1} \sum_i (a_i - \bar{a})(a'_i - \bar{a}') \sum_j (b_j - \bar{b})^2. \tag{2}$$

Hoeffding's combinatorial central limit theorem [9, Thm. 3] gives asymptotic normality of $\sum_i c_N(i, \pi(i))$ when, for every integer $r \geq 3$, $N^{-1}\sum_{i,j} |d_N(i,j)|^r$ is $O\big((N^{-1}\sum_{i,j} d_N(i,j)^2)^{r/2}\big)$, d_N being the doubly centred array; for product arrays $c_N(i,j) = a_i b_j$ the condition reduces to a bound on the standardised absolute moments of a and b. In Section 5.2 the energies of a random arrangement are quadratic forms in such statistics; (2) gives their means exactly, and the distribution of a quadratic form in Gaussian variables can be computed by numerical inversion of its characteristic function, as in Imhof's method [10].

2.4 Enumeration and related polytopes

There are 8 magic squares of order 3, all images of the Lo Shu under the dihedral group D_4 of the square; 7,040 of order 4, forming 880 D_4-orbits first listed by Frénicle de Bessy and verified many

times since [3]; and 275,305,224 of order 5 [8]. For $n \geq 6$ only statistical estimates are known [21]. Classical constructions show that magic squares exist for all $n \geq 3$ [2].

The polytope literature concerns a different object from the one used here. An $n \times n$ nonnegative integer matrix with all row and column sums equal to t is a lattice point of the t-th dilate of the Birkhoff polytope, and their number is a polynomial in t [20, 7, 23]; adding the diagonal conditions and distinct entries leads to the inside-out polytopes of Beck and Zaslavsky [4, 3]; the cones of magic squares and cubes have Hilbert-basis descriptions [1]; and contingency tables with fixed margins are the same lattice points studied as a sample space [6]. Spectral and matrix properties of a magic square [13, 15, 16], the singular-value signatures of Cameron, Rogers and Loly [5], and Loly's physical invariants [11, 18, 19, 12] are closer to our viewpoint: they attach a quantity to one square and ask what it reveals.

3 Orthogonality conditions on the height function

Throughout, $n \geq 3$, $N = n^2$, and a cell of the $n \times n$ grid is a pair $c = (i, j)$ with $0 \leq i, j \leq n-1$; an arrangement is a bijection $S = (s_{ij})$ from the cells onto $\{1, \ldots, N\}$ (Definition 1.1). We use the centred coordinates of Definition 2.1,

$$x_{ij} = j - \tfrac{n-1}{2}, \qquad y_{ij} = \tfrac{n-1}{2} - i, \qquad z_{ij} = s_{ij} - \tfrac{N+1}{2},$$

and refer to $z = z(S)$ as the *height function* of S. A *line* is one of the n rows, the n columns, the main diagonal $\{i = j\}$ or the anti-diagonal $\{i + j = n-1\}$; there are $2n+2$ lines, each containing exactly n cells. With $M(n) = n(N+1)/2$ the magic constant, the *deviation* of a line L is

$$\delta_L(S) \;=\; \sum_{c \in L} s_c - M(n) \;=\; \sum_{c \in L} z_c, \tag{3}$$

the second equality because $|L| = n$ and $M(n) = n \cdot \frac{N+1}{2}$. S is *semi-magic* if $\delta_L = 0$ for the $2n$ rows and columns, and *magic* if $\delta_L = 0$ for all $2n+2$ lines. We write $\rho_k = \delta_{\text{row } k}$, $\gamma_\ell = \delta_{\text{col } \ell}$, δ_{m} and δ_{a} for the row, column, main-diagonal and anti-diagonal deviations.

3.1 Height functions, the inner product, and the energies

Let $\mathcal{F} = \mathbb{R}^{\text{cells}}$ be the N-dimensional space of real functions on the grid, with the normalised inner product and norm

$$\langle f, g \rangle = \frac{1}{N} \sum_{i,j} f_{ij} g_{ij}, \qquad \|f\|^2 = \langle f, f \rangle. \tag{4}$$

The mean of f is $\bar{f} = \langle f, \mathbf{1} \rangle$ and its centring is $\tilde{f} = f - \bar{f}\mathbf{1}$. For two functions ("test function" φ and height f) the empirical covariance over the N cells is $\mathrm{Cov}(\varphi, f) = \langle \tilde{\varphi}, \tilde{f} \rangle = \langle \tilde{\varphi}, f \rangle = \langle \varphi, \tilde{f} \rangle$; the last two equalities hold because $\tilde{\varphi} \perp \mathbf{1}$ and $\tilde{f} \perp \mathbf{1}$.

Lemma 3.1 (Means, variances, and covariance as an inner product)**.** *For every arrangement S of $\{1, \ldots, N\}$:*

1. *$\bar{x} = \bar{y} = \bar{z} = 0$, so z is centred and $\mathrm{Cov}(\varphi, Z) = \langle \tilde{\varphi}, z \rangle = \langle \varphi, z \rangle$ for every $\varphi \in \mathcal{F}$;*
2. $\mathrm{Var}(X) = \mathrm{Var}(Y) = \dfrac{n^2 - 1}{12}, \quad \mathrm{Var}(Z) = \dfrac{N^2 - 1}{12} = \dfrac{n^4 - 1}{12}, \quad \mathrm{Cov}(X, Y) = 0$*;*
3. *for every line L with indicator $\mathbf{1}_L$,* $\mathrm{Cov}(\mathbf{1}_L, Z) = \dfrac{\delta_L(S)}{N}$.

Proof. (1) $\sum_{i,j} x_{ij} = n \sum_j (j - \frac{n-1}{2}) = 0$, likewise for y, and $\sum_{i,j} z_{ij} = \sum_{v=1}^{N} v - N\frac{N+1}{2} = 0$; then $\mathrm{Cov}(\varphi, Z) = \langle \varphi, z \rangle - \bar{\varphi}\langle \mathbf{1}, z \rangle = \langle \varphi, z \rangle$. (2) $\mathrm{Var}(X) = \frac{1}{n}\sum_j (j - \frac{n-1}{2})^2$ is the variance of the uniform distribution on n equally spaced points, $(n^2 - 1)/12$; the same for Y, and for Z with N points. $\sum_{i,j} x_{ij} y_{ij} = (\sum_j x_{\cdot j})(\sum_i y_{i\cdot}) = 0$. (3) By (1), $\mathrm{Cov}(\mathbf{1}_L, Z) = \frac{1}{N}\sum_{c \in L} z_c = \delta_L / N$ by (3). □

We write $R_k = \mathbf{1}_{\{i=k\}}$, $C_\ell = \mathbf{1}_{\{j=\ell\}}$, $D_{\mathrm{m}} = \mathbf{1}_{\{i=j\}}$, $D_{\mathrm{a}} = \mathbf{1}_{\{i+j=n-1\}}$ for the $2n+2$ line indicators.

Definition 3.2 (Complete and low-mode energies)**.** For an arrangement S with height function z,

$$E_{\mathrm{full}}(S) = \sum_{k=0}^{n-1} \mathrm{Cov}(R_k, Z)^2 + \sum_{\ell=0}^{n-1} \mathrm{Cov}(C_\ell, Z)^2 + \mathrm{Cov}(D_{\mathrm{m}}, Z)^2 + \mathrm{Cov}(D_{\mathrm{a}}, Z)^2, \tag{5}$$

$$E_{\mathrm{low}}(S) = \mathrm{Cov}(X, Z)^2 + \mathrm{Cov}(Y, Z)^2 + \mathrm{Cov}(D_{\mathrm{m}}, Z)^2 + \mathrm{Cov}(D_{\mathrm{a}}, Z)^2. \tag{6}$$

Note that (5) sums over *all* n rows and *all* n columns. The word "energy" is a label for these sums of squares and nothing more: no Hamiltonian is implied, and the temperature of the annealing in Section 6.2 is a search parameter only. By Lemma 3.1(3) it has the following elementary description, which we use constantly:

$$N^2 E_{\mathrm{full}}(S) = \sum_L \delta_L(S)^2 = \sum_k \rho_k^2 + \sum_\ell \gamma_\ell^2 + \delta_{\mathrm{m}}^2 + \delta_{\mathrm{a}}^2, \tag{7}$$

the sum of the squared line-sum deviations over all $2n + 2$ lines. In particular $N^2 E_{\mathrm{full}}(S)$ is a non-negative integer.

Remark 3.3 (The four-term energy does not characterise magic squares for $n \geq 4$). $E_{\mathrm{low}} = 0$ imposes four linear conditions on the centred height functions, while "magic" imposes $2n$ independent ones (Lemma 3.6); for $n \geq 4$ the two cannot agree even before the permutation constraint is taken into account. A witness for $n = 4$ is

$$S = \begin{pmatrix} 9 & 4 & 12 & 3 \\ 16 & 1 & 14 & 7 \\ 10 & 15 & 11 & 8 \\ 2 & 5 & 6 & 13 \end{pmatrix},$$

whose diagonals both sum to 34 while the row sums are $(28, 38, 44, 26)$ and the column sums $(37, 25, 43, 31)$; nevertheless $\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = 0$, so $E_{\mathrm{low}}(S) = 0$ although $E_{\mathrm{full}}(S) = 396/256$. Proposition 5.4 counts all such arrangements, and Remark 3.17 explains why none exist for $n = 3$.

The dihedral group D_4 acts on the grid by the rotations and reflections of the square, on arrangements by $(gS)_c = s_{g^{-1}c}$ and on functions by $(gf)_c = f_{g^{-1}c}$, so that $z(gS) = g\, z(S)$ and $\langle gf, gh\rangle = \langle f, h\rangle$. Every g permutes the $2n$ rows and columns among themselves and the two diagonals among themselves, and acts on the centred coordinates by a signed permutation: $\{gx, gy\} = \{\pm x, \pm y\}$ as unordered pairs up to sign.

Proposition 3.4 (D_4-invariance of the energies)**.** *For every $g \in D_4$ and every arrangement S: $E_{\mathrm{full}}(gS) = E_{\mathrm{full}}(S)$ and $E_{\mathrm{low}}(gS) = E_{\mathrm{low}}(S)$.*

Proof. For any line L, $\delta_{gL}(gS) = \sum_{c \in gL} s_{g^{-1}c} = \delta_L(S)$, and $L \mapsto gL$ is a bijection of the set of lines, so the multiset of deviations is invariant and (7) gives $E_{\mathrm{full}}(gS) = E_{\mathrm{full}}(S)$. For E_{low}, $\mathrm{Cov}(X, gZ) = \langle g^{-1}x, z\rangle$ with $g^{-1}x \in \{\pm x, \pm y\}$, and similarly for y, so $\mathrm{Cov}(X, gZ)^2 + \mathrm{Cov}(Y, gZ)^2$ is invariant; the diagonal terms are permuted among themselves. □

Remark 3.5 (Why all $2n + 2$ lines). Since $\sum_k \rho_k = \sum_\ell \gamma_\ell = 0$, the last row and the last column are redundant for the zero set. They are not redundant for the values: dropping them would give a function that is not D_4-invariant, because the reflection that maps column $n - 1$ to column 0 would change it by $(\gamma_{n-1}^2 - \gamma_0^2)/N^2$. Keeping all $2n + 2$ terms (two more than the dimension $2n$ of the constraint space, Lemma 3.6) is the price of invariance, and all statistics of E_{full} below refer to (5).

3.2 The projection lemma

Let $\tilde{R}_k = R_k - \frac{1}{n}\mathbf{1}$, $\tilde{C}_\ell = C_\ell - \frac{1}{n}\mathbf{1}$, $\tilde{D}_{\mathrm{m}} = D_{\mathrm{m}} - \frac{1}{n}\mathbf{1}$, $\tilde{D}_{\mathrm{a}} = D_{\mathrm{a}} - \frac{1}{n}\mathbf{1}$ be the centred line indicators (each line has n of the N cells, so each mean is $1/n$), and set

$$W_R = \operatorname{span}\{\tilde{R}_k\}, \quad W_C = \operatorname{span}\{\tilde{C}_\ell\}, \quad W_D = \operatorname{span}\{\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}}\}, \quad W = W_R + W_C + W_D \subset \mathbf{1}^{\perp}.$$

Lemma 3.6 (Structure of W)**.** *For $n \geq 3$:*

1. *$W_R = \{f : \ f_{ij} = a_i, \ \sum_i a_i = 0\}$ and $W_C = \{f : \ f_{ij} = b_j, \ \sum_j b_j = 0\}$; both have dimension $n-1$.*
2. *$\langle \tilde{R}_k, \tilde{R}_{k'} \rangle = \frac{1}{N}(n\delta_{kk'} - 1)$, and the same for columns.*
3. *$W_R \perp W_C$, $W_R \perp W_D$, $W_C \perp W_D$.*
4. *$\langle \tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{m}} \rangle = \langle \tilde{D}_{\mathrm{a}}, \tilde{D}_{\mathrm{a}} \rangle = \frac{n-1}{N}$ and $\langle \tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}} \rangle = \frac{\epsilon_n}{N}$ with $\epsilon_n = 0$ for n odd and $\epsilon_n = -1$ for n even; in particular $\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}}$ are linearly independent and $\dim W_D = 2$.*
5. *$W = W_R \oplus W_C \oplus W_D$ is an orthogonal direct sum and $\dim W = 2n$.*

Proof. For two lines L, L', $N\langle \tilde{\mathbf{1}}_L, \tilde{\mathbf{1}}_{L'} \rangle = |L \cap L'| - 1$, since each line has n of the $N = n^2$ cells. Two distinct rows are disjoint and a row meets itself in n cells (2); a row meets every column and each diagonal in exactly one cell, and so does a column (3); the two diagonals meet in the centre cell when n is odd and not at all when n is even (4), and the Gram matrix $\frac{1}{N}\left(\begin{smallmatrix} n-1 & \epsilon_n \\ \epsilon_n & n-1 \end{smallmatrix}\right)$ has positive determinant for $n \geq 3$. For (1), $\sum_k a_k \tilde{R}_k$ is the function $(i,j) \mapsto a_i - \bar{a}$, and every zero-mean function of i alone arises this way; the map $a \mapsto a - \bar{a}$ has rank $n-1$. (5) follows: the three subspaces are mutually orthogonal, hence independent, and $\dim W = 2(n-1) + 2$. □

For a subspace $V \subset \mathcal{F}$ let P_V denote the orthogonal projection with respect to (4).

Lemma 3.7 (Projection lemma)**.** *Let S be an arrangement with height function z and line deviations $\rho, \gamma \in \mathbb{R}^n$, $\delta = (\delta_{\mathrm{m}}, \delta_{\mathrm{a}}) \in \mathbb{R}^2$, and let $G = \left(\begin{smallmatrix} n-1 & \epsilon_n \\ \epsilon_n & n-1 \end{smallmatrix}\right)$ as in Lemma 3.6(4). Then*

1. *$P_{W_R} z$ is the function $(i,j) \mapsto \rho_i/n$ (the row means of z) and $P_{W_C} z$ is $(i,j) \mapsto \gamma_j/n$, with $\sum_k \operatorname{Cov}(R_k, Z)^2 = \|\rho\|^2/N^2 = \frac{1}{n}\|P_{W_R} z\|^2$ and likewise for columns;*
2. *$\|P_{W_D} z\|^2 = \frac{1}{N}\,\delta^{\mathsf{T}} G^{-1} \delta$, while $\operatorname{Cov}(D_{\mathrm{m}}, Z)^2 + \operatorname{Cov}(D_{\mathrm{a}}, Z)^2 = \delta^{\mathsf{T}}\delta/N^2$;*
3. *consequently*

$$E_{\mathrm{full}}(S) = \frac{1}{n}\Big(\|P_{W_R} z\|^2 + \|P_{W_C} z\|^2\Big) + \frac{\delta^{\mathsf{T}}\delta}{N^2} = Q\big(P_W z\big), \tag{8}$$

where $Q(w) = \frac{1}{n}\|P_{W_R} w\|^2 + \frac{1}{n}\|P_{W_C} w\|^2 + \frac{1}{N^2} w_D^{\mathsf{T}} G^2 w_D$ (w_D the coordinates of $P_{W_D} w$ in the basis $\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}}$) is a positive-definite quadratic form on W. In particular E_{full} depends on z only through $P_W z$; but $E_{\mathrm{full}} \neq \|P_W z\|^2$ in general, because the centred indicators are not an orthonormal system.

Proof. (1) The orthogonal projection of z onto the functions of i alone is the function of row means $(i,j) \mapsto \rho_i/n$, which has zero mean because z does, so it is $P_{W_R} z$; its squared norm is $\|\rho\|^2/n^3$ while $\sum_k \operatorname{Cov}(R_k, Z)^2 = \|\rho\|^2/n^4$ by Lemma 3.1(3). Columns are identical. (2) The Gram matrix of $(\tilde{D}_{\mathrm{m}}, \tilde{D}_{\mathrm{a}})$ is G/N and the vector of inner products with z is δ/N, so $\|P_{W_D} z\|^2 = (\delta/N)^{\mathsf{T}}(G/N)^{-1}(\delta/N) = \frac{1}{N}\delta^{\mathsf{T}} G^{-1}\delta$. (3) Sum (1) and (2). If $P_{W_D} z = w_{\mathrm{m}}\tilde{D}_{\mathrm{m}} + w_{\mathrm{a}}\tilde{D}_{\mathrm{a}}$ then $\delta = G w_D$, so $\delta^{\mathsf{T}}\delta/N^2 = w_D^{\mathsf{T}} G^2 w_D/N^2$; each piece of Q is positive definite on its orthogonal summand (G is invertible), so Q is positive definite on W. □

Lemma 3.8 (Zero set of the energies)**.** *For every arrangement S of $\{1, \ldots, N\}$:*

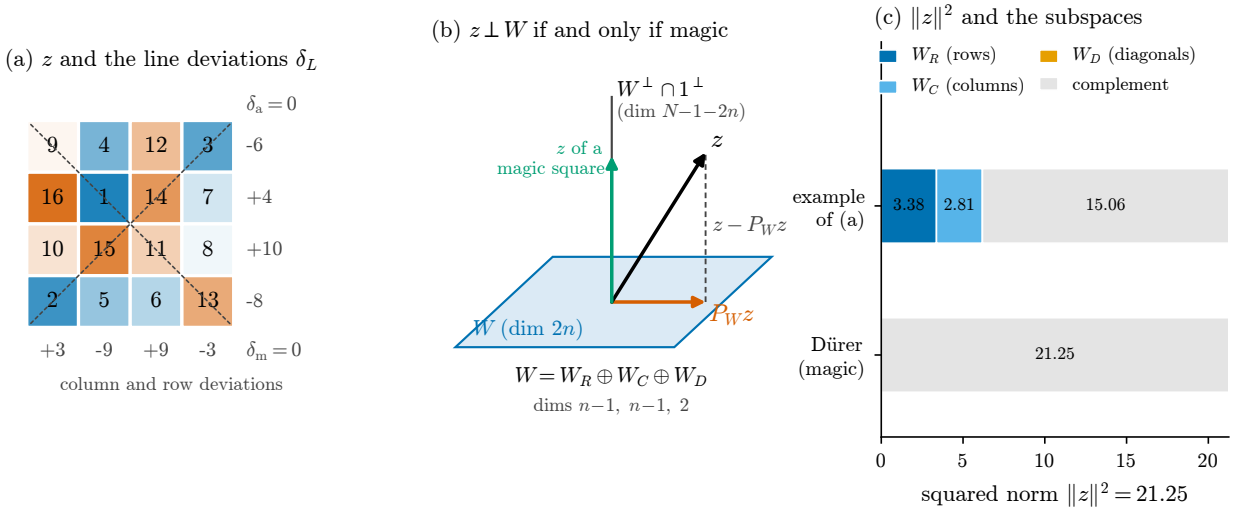


Figure 1: The projection lemma for $n = 4$. (a) The arrangement of Remark 3.3, shaded by the height z, with its row and column deviations (the diagonal sums are both 34). (b) Schematic: z splits orthogonally into $P_W z \in W$ and a component in $W^\perp \cap \mathbf{1}^\perp$; the arrangement is magic if and only if the first component vanishes. (c) Squared norms $\|P_{W_R} z\|^2, \|P_{W_C} z\|^2, \|P_{W_D} z\|^2$ and that of the complement, for the square of (a) $(3.38 + 2.81 + 0 + 15.06)$ and for Dürer's magic square (all of $\|z\|^2 = 21.25$ lies in the complement).

1. $E_{\text{full}}(S) = 0 \iff P_W z = 0 \iff z \perp W \iff S$ *is a magic square.*
2. $\sum_k \operatorname{Cov}(R_k, Z)^2 + \sum_\ell \operatorname{Cov}(C_\ell, Z)^2 = 0 \iff z \perp W_R \oplus W_C \iff S$ *is semi-magic.*
3. $E_{\text{low}}(S) = 0 \implies$ *both diagonals sum to* $M(n)$*; the implication "*$E_{\text{low}}(S) = 0 \implies S$ *magic" holds for* $n = 3$ *and fails for* $n = 4$ *(Remark 3.17, Proposition 5.4).*

Proof. (1) $E_{\text{full}}(S) = 0$ iff $Q(P_W z) = 0$ iff $P_W z = 0$ iff $z \perp W$, and by (7) iff all line deviations vanish. (2) is the same with the diagonal terms removed; (3) is immediate from (6) and Remark 3.17. □

Remark 3.9 (What the projection lemma does and does not say). Lemma 3.8(1) is the definition of a magic square rewritten in the language of orthogonality; it is not a theorem about magic squares, and we do not present it as one. Its use is as a change of viewpoint, which makes three things available that are not visible in the line-sum formulation: (i) W can be replaced by polynomial test functions (Section 3.3), which produces a hierarchy of relaxations whose degree-two level implies Loly's inertia invariance, and orthogonality to W is a balance condition on the signed mass z (Section 4); (ii) for a uniformly random arrangement the quantities $\langle \varphi, z \rangle$ are permutation statistics in the sense of Hoeffding, with computable moments and limit law (Section 5.2); (iii) a single swap changes z by a multiple of a difference of two cell indicators, and (7) turns the change of E_{full} into a two-line count (Section 3.4).

3.3 The moment hierarchy

The functions x and y are the degree-one polynomials in the column and row index. Replacing them by higher powers gives a natural family of relaxations of the semi-magic condition.

Definition 3.10 (Degree-k conditions)**.** Let $f \in \mathcal{F}$ be centred ($\langle f, \mathbf{1} \rangle = 0$) and $k \geq 1$. We say f satisfies the *degree-k conditions* if

$$\langle x^m, f \rangle = 0 \quad \text{and} \quad \langle y^m, f \rangle = 0 \qquad \text{for all } 1 \leq m \leq k.$$

For $f = z(S)$ these read $\operatorname{Cov}(X^m, Z) = \operatorname{Cov}(Y^m, Z) = 0$ for $1 \leq m \leq k$, by Lemma 3.1(1).

Write $x_j = j - \frac{n-1}{2}$ and $y_i = \frac{n-1}{2} - i$ for the n distinct values taken by x and y, and for centred f let $\gamma_j(f) = \sum_i f_{ij}$ and $\rho_i(f) = \sum_j f_{ij}$ be its column and row sums (for $f = z(S)$ these are the deviations γ, ρ of (3)).

Proposition 3.11 (Moment hierarchy)**.** *Let $n \geq 3$ and let $f \in \mathcal{F}$ be centred.*

(a) For every function φ that depends only on the column index, $\langle \varphi, f\rangle = \frac{1}{N}\sum_j \varphi(j)\gamma_j(f)$; in particular $\langle x^m, f\rangle = \frac{1}{N}\sum_j x_j^m \gamma_j(f)$ and $\langle y^m, f\rangle = \frac{1}{N}\sum_i y_i^m \rho_i(f)$ for all $m \geq 0$. Hence if $\gamma(f) = 0$ and $\rho(f) = 0$ (for $f = z(S)$: if S is semi-magic), then $\langle x^m, f\rangle = \langle y^m, f\rangle = 0$ for every $m \geq 0$.

(b) If $k \geq n-1$, the degree-k conditions imply $\gamma(f) = \rho(f) = 0$.

(c) If $1 \leq k \leq n-2$, there is a centred f satisfying the degree-k conditions with $\gamma(f) \neq 0$; explicitly, $f_{ij} = \frac{1}{n}(-1)^j\binom{k+1}{j}$ for $0 \leq j \leq k+1$ and $f_{ij} = 0$ for $j > k+1$. This f violates the degree-$(k+1)$ condition.

Thus, for real-valued height functions, the degree-k conditions are equivalent to "equal row sums and equal column sums" exactly when $k \geq n-1$.

Parts (b) and (c) are the invertibility of the Vandermonde matrix of the nodes $x_0, \ldots, x_{n-1}$, and we claim no novelty for them; the threshold $n-1$ is the benchmark against which Theorem 3.15 is measured. Part (a) is, in physical language, the vanishing of the axial moments $\sum_{ij} z_{ij} x_j^m$ of the signed charge distribution z of a semi-magic square (Remark 4.6), in the spirit of Rogers and Loly's multipole analysis of a magic cube [19]. Note that the hierarchy uses the pure powers x^m, y^m only: the mixed moments $\sum_{ij} s_{ij} x_j^a y_i^b$ with $a, b \geq 1$, beginning with the product of inertia $\sum_{ij} s_{ij} x_j y_i$, are not determined by the line sums (Remark 3.22).

Proof. (a) $\langle \varphi, f\rangle = \frac{1}{N}\sum_j \varphi(j)\sum_i f_{ij} = \frac{1}{N}\sum_j \varphi(j)\gamma_j(f)$; the row statement is symmetric. The last sentence is immediate.

(b) Suppose $\langle x^m, f\rangle = 0$ for $1 \leq m \leq k$ with $k \geq n-1$. Together with $\sum_j \gamma_j(f) = \sum_{i,j} f_{ij} = 0$ (the case $m = 0$), part (a) says that $\gamma(f) \in \mathbb{R}^n$ is orthogonal to the n vectors $(x_j^m)_j$, $m = 0, \ldots, n-1$. These are the columns of the Vandermonde matrix of the n *distinct* nodes $x_0, \ldots, x_{n-1}$, which is invertible; hence they span $\mathbb{R}^n$ and $\gamma(f) = 0$. The same argument with the nodes y_i gives $\rho(f) = 0$.

(c) Let $v_j = (-1)^j\binom{k+1}{j}$ for $0 \leq j \leq k+1$ and $v_j = 0$ for $k+1 < j \leq n-1$ (this needs $k+1 \leq n-1$). For a polynomial p define the forward difference $(\Delta p)(t) = p(t+1) - p(t)$; Δ lowers the degree of a nonconstant polynomial by exactly one and kills constants, so $\Delta^{k+1}p = 0$ whenever $\deg p \leq k$. Writing E for the shift $(Ep)(t) = p(t+1)$ we have $\Delta = E - 1$, and since E and 1 commute the binomial theorem gives $\Delta^{k+1} = \sum_{j=0}^{k+1}\binom{k+1}{j}(-1)^{k+1-j}E^j$, i.e.

$$(\Delta^{k+1}p)(0) = (-1)^{k+1}\sum_{j=0}^{k+1}(-1)^j\binom{k+1}{j}p(j) = (-1)^{k+1}\sum_{j=0}^{n-1} v_j\, p(j). \tag{9}$$

Taking $p(t) = (t - \frac{n-1}{2})^m$ with $0 \leq m \leq k$ shows $\sum_j v_j x_j^m = 0$; in particular ($m = 0$) $\sum_j v_j = 0$. Now set $f_{ij} = v_j/n$. Then $\sum_{i,j} f_{ij} = \sum_j v_j = 0$, so f is centred; $\rho_i(f) = \frac{1}{n}\sum_j v_j = 0$ for every i, so all $\langle y^m, f\rangle$ vanish by (a); $\gamma_j(f) = v_j$, so $\langle x^m, f\rangle = \frac{1}{N}\sum_j v_j x_j^m = 0$ for $1 \leq m \leq k$ by (9); and $\gamma_0(f) = v_0 = 1 \neq 0$. Finally, for $p(t) = (t - \frac{n-1}{2})^{k+1}$ we have $\Delta^{k+1}p \equiv (k+1)! \neq 0$, so $\langle x^{k+1}, f\rangle = \pm(k+1)!/N \neq 0$. □

Corollary 3.12 (E_{low} is the degree-one truncation)**.** *$E_{\text{low}}(S) = \langle x, z\rangle^2 + \langle y, z\rangle^2 + (\delta_{\text{m}}^2 + \delta_{\text{a}}^2)/N^2$, so $E_{\text{low}}(S) = 0$ iff z satisfies the degree-1 conditions and both diagonals sum to $M(n)$ (Lemma 3.1). By Proposition 3.11(a) every magic square has $E_{\text{low}} = 0$; by Proposition 3.11(c) the degree-1 conditions characterise equal line sums among real height functions only when $n - 1 \leq 1$, i.e. never for $n \geq 3$.*

Corollary 3.13 (Loly's invariant from the hierarchy)**.** *Place a mass s_{ij} at the planar point (x_{ij}, y_{ij}) and let $I(S) = \sum_{i,j} s_{ij}\big(x_{ij}^2 + y_{ij}^2\big)$ be the moment of inertia about the centre [11]. Then for every arrangement*

$$I(S) = N\,\mathrm{Cov}\big(Z,\, X^2 + Y^2\big) + \frac{n^2(n^4-1)}{12} = \sum_j x_j^2\gamma_j + \sum_i y_i^2\rho_i + \frac{n^2(n^4-1)}{12}. \tag{10}$$

Hence $I(S)$ takes Loly's value $n^2(n^4-1)/12$ if and only if the single condition $\mathrm{Cov}(Z, X^2+Y^2) = 0$ holds. This condition is implied by the degree-2 conditions of Definition 3.10, and therefore (Proposition 3.11(a)) by the semi-magic property, but it is much weaker than the degree-2 conditions (see below). The two halves $I_x = \sum_j x_j^2(\gamma_j + M(n))$ and $I_y = \sum_i y_i^2(\rho_i + M(n))$ each equal $n^2(n^4-1)/24$ for semi-magic squares.

The invariance and its validity for semi-magic squares are Loly's [11]: his derivation writes $I = I_x + I_y$ with $I_x = \sum_j x_j^2 \cdot (\text{column sum } j)$, $I_y = \sum_i y_i^2 \cdot (\text{row sum } i)$, uses the row and column sums only, and gives $I_x = I_y = n^2(n^4-1)/24$. The corollary re-derives this; what it adds is the decomposition (10) into a degree-two cross-moment plus a constant, and the observation that the invariance is one scalar condition rather than the degree-two level. How much weaker the single condition is can be seen for $n = 3$ by enumeration of all 9! arrangements (`anc/verify_numbers.py`): 24,192 of them have $I = 60$, against 72 satisfying the degree-2 conditions (the semi-magic squares, Remark 3.17). Loly's value $n^2(n^4-1)/12$ equals 60, 340, 1300, 3885, 9800, 21840, 44280 for $n = 3, \ldots, 9$.

Proof. Substitute $s_{ij} = z_{ij} + \frac{N+1}{2}$: $I(S) = \sum_{i,j} z_{ij}(x_{ij}^2 + y_{ij}^2) + \frac{N+1}{2}\sum_{i,j}(x_{ij}^2 + y_{ij}^2)$. The first sum is $N\mathrm{Cov}(X^2+Y^2, Z)$ by Lemma 3.1(1) and equals $\sum_j x_j^2\gamma_j + \sum_i y_i^2\rho_i$ by Proposition 3.11(a); the second is $\frac{N+1}{2}\cdot 2\cdot\frac{n^2(n^2-1)}{12} = \frac{n^2(n^4-1)}{12}$ by Lemma 3.1(2). For semi-magic squares $\gamma = \rho = 0$ and $I_x = M(n)\sum_j x_j^2 = \frac{n(n^2+1)}{2}\cdot\frac{n(n^2-1)}{12}$. □

Corollary 3.14 (Null distribution of Loly's invariant)**.** *For a uniformly random arrangement S of order n,*

$$\mathbb{E}[I(S)] = \frac{n^2(n^4-1)}{12}, \qquad \mathrm{Var}\big(I(S)\big) = N^2\sigma_n^2\,\mathrm{Var}(X^2+Y^2) = \frac{n^4(n^4-1)(n^2-4)}{1080}, \tag{11}$$

where $\sigma_n^2 = (n^2+1)/12$ and $\mathrm{Var}(X^2+Y^2) = (n^2-1)(n^2-4)/90$ is the variance of x^2+y^2 over the N cells. The mean equals Loly's value for every distribution of arrangements that is invariant under relabelling of the values $1, \ldots, N$ (in particular the uniform distribution): magic and semi-magic squares sit exactly at the population mean of I, and the invariance says that their fluctuation vanishes, not that their value is extreme. For $n = 3$, $\mathrm{Var}(I) = 30$ (standard deviation 5.48; 29 values from 46 to 74); for $n = 4$, $\mathrm{Var}(I) = 2176/3$ (standard deviation 26.9), and 2.9% of arrangements have $I = 340$.

Proof. $N\mathrm{Cov}(Z, X^2+Y^2) = \sum_c \big(x_c^2 + y_c^2 - \overline{x^2+y^2}\big) z_{\pi(c)}$ is a linear permutation statistic with mean 0 (Section 2.3); by (2) with $a = x^2+y^2$ and $b = z$ its variance is $\frac{1}{N-1}\cdot N\mathrm{Var}(X^2+Y^2)\cdot N\mathrm{Var}(Z) = N^2\sigma_n^2\mathrm{Var}(X^2+Y^2)$, as in Proposition 5.1. Since X and Y are independent and uniform on n equally spaced points, $\mathrm{Var}(X^2+Y^2) = 2\mathrm{Var}(X^2) = 2(\mathbb{E}X^4 - (\mathbb{E}X^2)^2)$ with $\mathbb{E}X^2 = (n^2-1)/12$ and $\mathbb{E}X^4 = (n^2-1)(3n^2-7)/240$, which gives $(n^2-1)(n^2-4)/90$. The $n = 3$ values are confirmed by enumeration of all 9! arrangements, and the $n = 4$ values by exact enumeration over the assignments of values to the three radial shells of the grid (`anc/verify_numbers.py`). □

Proposition 3.11 is a statement about real-valued height functions. For arrangements, i.e. for $f = z(S)$ with S a permutation of $\{1, \ldots, N\}$, the integrality and boundedness of the line sums change the picture in an interesting way.

Theorem 3.15 (Integrality lowers the permutation threshold)**.** *For $n \geq 3$ let $k^*(n)$ be the least k such that every arrangement S of $\{1, \ldots, N\}$ satisfying the degree-k conditions is semi-magic. Then $k^*(n) \leq n-2$ for all $n \geq 14$: for such n the degree-$(n-2)$ conditions already force equal row and column sums. By contrast $k^*(n) \leq n-1$ for all n (Proposition 3.11(b)), with equality for $n = 3, 4$: $k^*(3) = 2$ and $k^*(4) = 3$ by the witnesses*

$$n = 3,\ \textit{degree } 1: \quad \begin{pmatrix} 1 & 3 & 9 \\ 8 & 6 & 5 \\ 7 & 4 & 2 \end{pmatrix}, \qquad \gamma = (1, -2, 1),\ \rho = (-2, 4, -2);$$

$$n = 4,\ \textit{degree } 2: \quad \begin{pmatrix} 14 & 2 & 5 & 13 \\ 8 & 15 & 7 & 4 \\ 12 & 3 & 9 & 10 \\ 1 & 11 & 16 & 6 \end{pmatrix}, \qquad \gamma = (1, -3, 3, -1),\ \rho = 0.$$

Proof. Lower bounds. For the 3×3 array, the column sums are $16, 13, 16$ and the row sums $13, 19, 13$, so $\gamma = (1, -2, 1)$, $\rho = (-2, 4, -2)$ and $\sum_j x_j \gamma_j = (-1)(1) + 0 + (1)(1) = 0$, $\sum_i y_i \rho_i = (1)(-2) + 0 + (-1)(-2) = 0$: the degree-1 conditions hold but the square is not semi-magic, so $k^*(3) \geq 2$. For the 4×4 array, every row sums to 34 and the column sums are $35, 31, 37, 33$, so $\rho = 0$ and $\gamma = (1, -3, 3, -1)$, which is the vector v of Proposition 3.11(c) with $k = 2$; hence $\sum_j x_j^m \gamma_j = 0$ for $m = 1, 2$ by (9), the degree-2 conditions hold, and the square is not semi-magic, so $k^*(4) \geq 3$.

The bound for $n \geq 14$. Let S satisfy the degree-$(n-2)$ conditions. By Proposition 3.11(a), γ is orthogonal to the $n-1$ linearly independent vectors $(x_j^m)_j$, $m = 0, \ldots, n-2$, so it lies in their one-dimensional orthogonal complement in $\mathbb{R}^n$. By (9) with $k+1 = n-1$, this complement is spanned by $v_j = (-1)^j \binom{n-1}{j}$, $j = 0, \ldots, n-1$ (which is orthogonal to all polynomials of degree $\leq n-2$ and nonzero). So $\gamma = tv$ with $t = \gamma_0 \in \mathbb{Z}$ because $v_0 = 1$ and γ is an integer vector. On the other hand every column sum of a permutation of $\{1, \ldots, N\}$ lies between the sum of the n smallest and the sum of the n largest values, so

$$|\gamma_j| \leq \sum_{v=N-n+1}^{N} v - M(n) = \frac{n(2N-n+1)}{2} - \frac{n(N+1)}{2} = \frac{n(N-n)}{2} = \frac{n^2(n-1)}{2}.$$

If $t \neq 0$ then $\max_j |\gamma_j| \geq \binom{n-1}{\lfloor (n-1)/2 \rfloor}$. We claim $\binom{n-1}{\lfloor (n-1)/2 \rfloor} > \frac{n^2(n-1)}{2}$ for all $n \geq 14$. For $n = 14$: $\binom{13}{6} = 1716 > 1274 = \frac{14^2 \cdot 13}{2}$. Passing from n to $n+1$ multiplies the left side by $\binom{n}{\lfloor n/2 \rfloor} / \binom{n-1}{\lfloor (n-1)/2 \rfloor}$, which equals 2 for n even and $\frac{2n}{n+1} \geq \frac{28}{15}$ for odd $n \geq 14$, while it multiplies the right side by $\frac{(n+1)^2 n}{n^2(n-1)} = \frac{(n+1)^2}{n(n-1)} \leq \frac{225}{182} < \frac{28}{15}$ for $n \geq 14$; the claim follows by induction. Hence $t = 0$ and $\gamma = 0$; the same argument applied to the rows gives $\rho = 0$. □

Remark 3.16 ($k^*(n) = n-1$ for $5 \leq n \leq 13$; computer-verified). For each $5 \leq n \leq 13$ a randomised swap search found an arrangement with all row sums equal to $M(n)$ and column deviations $\gamma_j = (-1)^j \binom{n-1}{j}$; by (9) such an arrangement satisfies the degree-$(n-2)$ conditions without being semi-magic, so $k^*(n) = n-1$ for all $3 \leq n \leq 13$. The $n = 5$ witness is printed in Section C; all nine are in the ancillary file `anc/data_witnesses_variances.json` and verified in exact integer arithmetic by `anc/verify_numbers.py`. Thus the permutation threshold agrees with the linear-algebra threshold exactly for $n \leq 13$ and drops below it from $n = 14$ on; determining $k^*(n)$ for $n \geq 14$ is open (Section 6).

Remark 3.17 (The low-mode energy for $n = 3$). Exhaustive enumeration of the 9! arrangements in exact arithmetic (`anc/verify_numbers.py`) gives 760 arrangements satisfying the degree-1 conditions, exactly 72 (the semi-magic squares) satisfying the degree-2 conditions, in accordance with $k^*(3) = 2$, and exactly 8 satisfying the degree-1 conditions together with both diagonal conditions—the magic squares. So for $n = 3$, $E_{\text{low}}(S) = 0 \iff S$ is magic, but for the "wrong" reason: the degree-1 conditions leave the families $\gamma = t(1, -2, 1)$, $\rho = u(1, -2, 1)$ open, and the two diagonal

conditions combined with the permutation constraint happen to force $t = u = 0$. Nothing of the sort survives for $n = 4$: the arrangement of Remark 3.3 has $E_{\text{low}} = 0$ with $\gamma = (3, -9, 9, -3)$, $\rho = (-6, 4, 10, -8)$; it satisfies the degree-2 condition in x but not in y ($\sum_i y_i^2 \rho_i = -28$).

3.4 The single-swap formula

Let $a \neq b$ be two cells and let S^{ab} be the arrangement obtained from S by exchanging the entries in a and b. Define

$$m(a,b) = \#\{\text{lines } L : \ L \text{ contains exactly one of } a, b\},$$
$$u_S(c) = \sum_{L \ni c} \delta_L(S) \qquad (\text{sum over the lines through } c).$$

Theorem 3.18 (Single-swap formula)**.** *For every arrangement S and cells $a \neq b$, with $d = s_b - s_a$,*

$$N^2\big(E_{\text{full}}(S^{ab}) - E_{\text{full}}(S)\big) = d^2\, m(a,b) + 2d\,\big(u_S(a) - u_S(b)\big). \tag{12}$$

In particular, if S is a magic square then, with $d = |s_a - s_b|$,

$$E_{\text{full}}(S^{ab}) = \frac{d^2\, m(a,b)}{N^2}. \tag{13}$$

Proof. The swap changes the height function by $z(S^{ab}) = z(S) + d(\mathbf{1}_a - \mathbf{1}_b)$. For a line L, $\delta_L(S^{ab}) = \delta_L(S) + d\big(\mathbf{1}_L(a) - \mathbf{1}_L(b)\big)$: the deviation increases by d if L contains a but not b, decreases by d if L contains b but not a, and is unchanged if L contains both or neither. By (7),

$$\begin{aligned} N^2 \Delta E_{\text{full}} &= \sum_{L \ni a,\, L \not\ni b} \big[(\delta_L + d)^2 - \delta_L^2\big] + \sum_{L \ni b,\, L \not\ni a} \big[(\delta_L - d)^2 - \delta_L^2\big] \\ &= d^2 m(a,b) + 2d\Big(\sum_{L \ni a,\, L \not\ni b} \delta_L - \sum_{L \ni b,\, L \not\ni a} \delta_L\Big). \end{aligned}$$

Adding and subtracting the lines containing both a and b turns the bracket into $u_S(a) - u_S(b)$. If S is magic, all $\delta_L(S) = 0$, so $u_S \equiv 0$ and $E_{\text{full}}(S) = 0$, giving (13). □

Lemma 3.19 (The multiplicity $m(a,b)$)**.** *For distinct cells $a = (i,j)$, $b = (i',j')$:*

$$\begin{aligned} m(a,b) = 2 \cdot [i \neq i'] + 2 \cdot [j \neq j'] &+ [\textit{exactly one of } a, b \textit{ on the main diagonal}] \\ &+ [\textit{exactly one of } a, b \textit{ on the anti-diagonal}]. \end{aligned}$$

Hence $2 \leq m(a,b) \leq 6$, and $m(a,b) = 2$ if and only if a and b lie in a common row or a common column and neither of them lies on a diagonal. Such pairs exist for every $n \geq 3$ (for $n = 3$ exactly one pair up to D_4: the two non-central cells of the middle row or column; for $n \geq 4$ any two off-diagonal cells of a row).

Proof. If $i = i'$ the common row contains both cells and no other row contains either; if $i \neq i'$ the row of a contains a only and the row of b contains b only; columns likewise, and each diagonal contributes 1 or 0 according to whether it separates a from b. Since $a \neq b$ the row/column contribution is 2 or 4, so $2 \leq m \leq 6$. If $m = 2$ then $i = i'$ or $j = j'$ (not both) and no diagonal separates the cells; for $i = i'$, $j \neq j'$ both cells cannot lie on the same diagonal, so "not separated" means neither is on it. For existence: a row contains at most two diagonal cells, so it has at least $n - 2 \geq 2$ off-diagonal cells when $n \geq 4$, and the middle row of the 3×3 grid has the two off-diagonal cells $(1,0), (1,2)$. □

Corollary 3.20 (Strict local minimality and the exact gap)**.** *Let S be a magic square of order $n \geq 3$.*

1. *Every single swap strictly increases the energy:* $E_{\text{full}}(S^{ab}) \geq 2/N^2 > 0$ *for all* $a \neq b$. *Thus every magic square is a strict local minimum of* E_{full} *on the set of arrangements with respect to transpositions, for every* n.

2. *The smallest energy reachable by one swap is*

$$\Delta_{\min}(S) = \min_{a \neq b} \frac{(s_a - s_b)^2\, m(a,b)}{N^2}, \qquad \frac{2}{N^2} \leq \Delta_{\min}(S) \leq \frac{6}{N^2}.$$

   The lower bound is attained iff two cells with consecutive values share a row or column and are both off the diagonals; the upper bound is the swap of any two consecutive values ($d = 1$, $m \leq 6$). *Thus* $\Delta_{\min}(S) = \Theta(n^{-4})$ *with explicit constants, uniformly over all magic squares of order* n.

3. *The possible values of* $N^2 E_{\text{full}}(S^{ab})$ *are the integers* $d^2 m$ *with* $1 \leq d \leq N-1$ *and* $2 \leq m \leq 6$. *For the* 880 *magic squares of order* 4 *the multiset of* $N^2\Delta_{\min}$ *is* $\{2 : 432,\ 3 : 420,\ 4 : 28\}$, *and for the* 8 *squares of order* 3 *it is* $\{3 : 8\}$ *(Section 5.3, Figure 4).*

Proof. By (13), $N^2 E_{\text{full}}(S^{ab}) = d^2 m(a,b)$ with $d \geq 1$ and $m \geq 2$ (Lemma 3.19); the cells carrying the values $v, v+1$ have $d = 1$ and $m \leq 6$, and the equality case of the lower bound is Lemma 3.19. □

Remark 3.21 (Swap formula for E_{low}). Since $\operatorname{Cov}(\varphi, Z^{ab}) = \operatorname{Cov}(\varphi, Z) + \frac{d}{N}(\varphi(a) - \varphi(b))$ for every test function φ (Lemma 3.1(1)), the same computation gives, for $E_{\text{low}}(S) = 0$ and in particular for S magic,

$$E_{\text{low}}(S^{ab}) = \frac{d^2}{N^2}\Big[(x_a - x_b)^2 + (y_a - y_b)^2 + \big(D_{\text{m}}(a) - D_{\text{m}}(b)\big)^2 + \big(D_{\text{a}}(a) - D_{\text{a}}(b)\big)^2\Big],$$

i.e. d^2/N^2 times the squared planar distance between the cells plus the number of diagonals separating them. The bracket is at least 1, so magic squares are strict local minima of E_{low} as well, with $E_{\text{low}}(S^{ab}) \geq 1/N^2$.

Remark 3.22 (Loly's invariant and the product of inertia). Loly's value-weighted planar moments of inertia have diagonal entries $\sum s_{ij} y_{ij}^2$ and $\sum s_{ij} x_{ij}^2$ that are fixed by the row and column sums (Corollary 3.13), whereas the product of inertia $P_{xy} = -\sum_{ij} s_{ij} x_{ij} y_{ij}$ is a mixed moment, fixed by no line sum: D_4 changes at most its sign, and over the 880 essentially different magic squares of order 4 the invariant $|P_{xy}|$ takes 48 distinct values between 0 and 60, vanishing for 56 of them. This is the reason the hierarchy of Definition 3.10 uses only the pure powers x^m, y^m.

4 Balance of the signed mass

Read the height z_c of a cell as a *signed mass* sitting at the planar position (x_c, y_c), or as the vertical coordinate of the point (x_c, y_c, z_c). A weighted configuration is *balanced* for an assignment of vectors u_c to the cells if the vector sum $B = \sum_c z_c\, u_c$ vanishes. The orthogonality formulation decides which assignments are balanced by every magic square (those whose components lie in the line space, Theorem 4.1), gives a quantitative version for near-magic arrangements (Theorem 4.2), and shows that the natural nonlinear assignments by unit vectors are not (Proposition 4.3). Table 1 summarises three physical readings and which balance conditions each yields.

Let $\mathcal{L} = \mathbb{R}\mathbf{1} \oplus W$ denote the span of the constant function and the $2n+2$ line indicators (Lemma 3.6); $\dim \mathcal{L} = 2n+1$. Recall $\langle f, g\rangle = \frac{1}{N}\sum_c f_c g_c$, so that $\sum_c z_c \varphi_c = N\langle z, \varphi\rangle$.

4.1 Linear balance

Theorem 4.1 (Linear balance)**.** *Let* S *be an arrangement of order* $n \geq 3$ *with height function* z.

(a) $E_{\text{full}}(S) = 0$ *if and only if* $\sum_c z_c \varphi_c = 0$ *for every* $\varphi \in \mathcal{L}$.

reading	configuration	implied by $E_{\text{full}} = 0$	not implied
flat plate (Loly): mass s_c at (x_c, y_c)	planar masses, centred grid	total mass $\sum s_c$ (fixed); centre of mass at the origin; axial moments $\sum s_c x_c^k$, $\sum s_c y_c^k$; moment of inertia $I = \sum s_c(x_c^2 + y_c^2) = \frac{n^2(n^4-1)}{12}$ (semi-magic suffices)	mixed moments, e.g. the product of inertia $\sum s_c x_c y_c$ (Theorem 4.1(d))
ring / top: weight z_c at angle θ_c of cell c	points on the unit circle, $u_c = (\cos\theta_c, \sin\theta_c)$	only the total weight $\sum z_c = 0$	$B_{\text{ring}} = 0$ ($B_{\text{ring}} = 2.62$ for all eight 3×3 magic squares)
vectors from the centroid, signed mass z_c	$v_c = (x_c, y_c, z_c)$; linear sums $\sum z_c v_c$ and unit-vector sums $\sum z_c v_c/\lvert v_c\rvert$	linear: horizontal part $\sum z_c(x_c, y_c) = 0$ and all separable moments; the vertical part $\sum z_c^2 = N\mathrm{Var}Z$ is fixed and positive	unit-vector sum: horizontal part (Proposition 4.3(a)); the vertical part $V > 0$ for every arrangement

Table 1: Three readings of a magic square as a weighted configuration, and the balance conditions that follow from $E_{\text{full}} = 0$. The plate reading uses the planar masses s_c; the other two use the signed heights z_c.

(b) If S is magic, then for all functions $a, b : \{0, \dots, n-1\} \to \mathbb{R}$ and all $c_1, c_2 \in \mathbb{R}$,

$$\sum_{i,j} z_{ij}\Big(a(x_{ij}) + b(y_{ij}) + c_1 \mathbf{1}_{D_{\text{m}}}(i,j) + c_2 \mathbf{1}_{D_{\text{a}}}(i,j)\Big) = 0. \tag{14}$$

In particular $\sum_c z_c = 0$ (true for every arrangement), the centre of mass vanishes, $\sum_c z_c(x_c, y_c) = 0$, every axial moment vanishes, $\sum_c z_c x_c^k = \sum_c z_c y_c^k = 0$ for all $k \geq 0$, and Loly's invariant holds, $\sum_c z_c(x_c^2 + y_c^2) = 0$ (Corollary 3.13). For the physical mass $s_c = z_c + \frac{N+1}{2}$ the centre of mass is the origin and every axial moment equals that of a uniform plate of the same total mass.

(c) For (b) with $c_1 = c_2 = 0$ it suffices that S is semi-magic.

(d) The mixed moments are not determined: for the magic square

$$\begin{pmatrix} 1 & 6 & 11 & 16 \\ 8 & 15 & 2 & 9 \\ 12 & 3 & 14 & 5 \\ 13 & 10 & 7 & 4 \end{pmatrix}$$

one has $\sum_c z_c x_c y_c = 60$, and $\sum_c z_c x_c y_c \neq 0$ for 824 of the 880 essentially different magic squares of order 4 (the same holds for $\sum_c z_c x_c^2 y_c$ and $\sum_c z_c x_c y_c^2$ for 623 and 773 of them). For $n = 3$ it vanishes for all eight magic squares.

Proof. (a) $\mathbf{1}^\perp \ni z$ for every arrangement (Lemma 3.1), so $z \perp \mathcal{L}$ if and only if $z \perp W$, which by Lemma 3.8 is equivalent to $E_{\text{full}}(S) = 0$. (b) $a(x) = \sum_\ell a(x_\ell) C_\ell$ lies in the span of the column indicators, $b(y)$ in that of the row indicators, and the diagonal indicators lie in $W + \mathbb{R}\mathbf{1}$; apply (a). The special cases are $a = 1$; $a(x) = x^k$, $b(y) = y^k$; and $a(x) = x^2$, $b(y) = y^2$. For the statement about s: $\sum_c s_c(x_c, y_c) = \sum_c z_c(x_c, y_c) + \frac{N+1}{2}\sum_c(x_c, y_c) = 0$, and $\sum_c s_c x_c^k = \frac{N+1}{2}\sum_c x_c^k$. (c) is Proposition 3.11(a). (d) The displayed square has all line sums 34; the counts for 880 squares are computed by enumeration. For $n = 3$ every magic square has $s_c + s_{c'} = 10$ for centrally opposite cells c, c', so z is odd under $(x, y) \mapsto (-x, -y)$ while xy is even. □

The linear moments of low degree therefore carry no information beyond $E_{\text{full}} = 0$, and one cannot hope for more from line sums alone: since $xy \notin \mathcal{L}$ (part (d) exhibits magic squares with $\sum_c z_c x_c y_c \neq 0$), the mixed moments vary over magic squares.

Theorem 4.2 (Quantitative linear balance)**.** *Let S be any arrangement of order $n \geq 3$ and let $E_{\mathrm{rc}} = \sum_k \mathrm{Cov}(R_k, Z)^2 + \sum_\ell \mathrm{Cov}(C_\ell, Z)^2$ $(\leq E_{\mathrm{full}})$ be the energy of the rows and columns.*

(a) $\mathrm{Cov}(X, Z)^2 + \mathrm{Cov}(Y, Z)^2 \;\leq\; \dfrac{n(n^2-1)}{12}\, E_{\mathrm{rc}}$, *equivalently*

$$\Big|\sum_c z_c(x_c, y_c)\Big|^2 \;\leq\; N^2\, \frac{n(n^2-1)}{12}\, E_{\mathrm{rc}} \;\leq\; N^2\, \frac{n(n^2-1)}{12}\, E_{\mathrm{full}}. \tag{15}$$

(b) The constant $n(n^2-1)/12$ cannot be improved, for any n: equality holds for the natural arrangement $s_{ij} = in+j+1$ (and, more generally, for every real height function $z = \alpha x + \beta y$), with equality throughout (15)*.*

(c) For every $\varphi \in W$, $\big|\sum_c z_c \varphi_c\big| \leq N\, \|\varphi\|\, \dfrac{n}{\sqrt{n-2}} \sqrt{E_{\mathrm{full}}}$, where $\|\varphi\|^2 = \frac{1}{N}\sum_c \varphi_c^2$.

Proof. (a) With $\gamma_\ell = \sum_i z_{i\ell}$ the column sums and ρ_k the row sums, $\sum_c z_c x_c = \sum_\ell x_\ell \gamma_\ell$ and $\sum_c z_c y_c = \sum_k y_k \rho_k$, where $\sum_\ell x_\ell^2 = \sum_k y_k^2 = n(n^2-1)/12$. Cauchy–Schwarz gives $(\sum_c z_c x_c)^2 + (\sum_c z_c y_c)^2 \leq \frac{n(n^2-1)}{12}(\|\gamma\|^2 + \|\rho\|^2)$, and $\|\gamma\|^2 + \|\rho\|^2 = N^2 E_{\mathrm{rc}}$ by Lemma 3.1(3). Dividing by N^2 gives the covariance form. (b) For the natural arrangement $z = x - ny$ the column sums are $\gamma = nx$ and the row sums $\rho = -n^2 y$, proportional to x and y, which is the equality case of both Cauchy–Schwarz steps; both diagonal sums vanish, so $E_{\mathrm{rc}} = E_{\mathrm{full}}$. In general equality in (a) holds iff $\gamma \parallel x$ and $\rho \parallel y$. (c) By Lemma 3.7, $\|P_{W_R} z\|^2 = n\|\rho\|^2/N^2$, $\|P_{W_C} z\|^2 = n\|\gamma\|^2/N^2$ and $\|P_{W_D} z\|^2 = \frac{1}{N}\delta^{\mathsf{T}} G^{-1} \delta$ with $\delta^{\mathsf{T}}\delta/N^2$ the diagonal part of E_{full}. The eigenvalues of G are $n-1$ (twice) for odd n and $n-2$, n for even n, so $\frac{1}{N}\delta^{\mathsf{T}} G^{-1}\delta \leq \frac{N}{n-2}\,\delta^{\mathsf{T}}\delta/N^2$, and $n\, E_{\mathrm{full}} \leq \|P_W z\|^2 \leq \max\{n, \frac{N}{n-2}\} E_{\mathrm{full}} = \frac{n^2}{n-2} E_{\mathrm{full}}$. For $\varphi \in W$, $\sum_c z_c \varphi_c = N\langle P_W z, \varphi\rangle \leq N\|\varphi\|\|P_W z\|$. □

Part (a) is a stability statement: the linear imbalance of the centre of mass is controlled by the energy, with a constant that depends only on n and is attained; for $E_{\mathrm{full}} = 0$ it recovers the centre-of-mass part of Theorem 4.1. Numerically, inequality (15) has no violation on all 9! arrangements for $n = 3$ (the ratio of the two sides reaches 1, at 4152 arrangements including the natural one), on 3×10^5 random arrangements for $n = 4$ (maximal ratio 0.998) and on 10^5 random arrangements for each of $n = 5, 6, 7$ (maximal ratios 0.977, 0.944, 0.903); see Figure 2(a). In (c), the observed ratio $\|P_W z\|^2 / E_{\mathrm{full}}$ over 2×10^4 random arrangements lies in $[n,\, 7.38]$ for $n = 4$, consistent with $[n,\, n^2/(n-2)] = [4, 8]$.

4.2 Nonlinear balance by unit vectors

Two assignments of unit vectors are natural. In the *ring* reading each cell $c \neq$ centre is projected radially onto the unit circle, $u_c = (x_c, y_c)/r_c$ with $r_c = \sqrt{x_c^2 + y_c^2}$ (and $u_c = 0$ at the centre cell when n is odd), and carries the weight z_c:

$$B_{\mathrm{ring}}(S) = \Big|\sum_c z_c\, u_c\Big|.$$

In the *three-dimensional* reading the vectors $v_c = (x_c, y_c, z_c)$ from the centroid are normalised, $u'_c = v_c/|v_c|$ (and $u'_c = 0$ if $v_c = 0$), and carry the weight z_c. Write B^{xy}_{3D} for the horizontal part of $\sum_c z_c u'_c$ and $V(S) = \sum_c z_c^2/|v_c|$ for its vertical part. Both are invariant under D_4, because $z(gS) = g\, z(S)$ and $u_{gc} = g\, u_c$.

Proposition 4.3 (Unit-vector balance is not implied)**.** *(a) Each of the eight magic squares of order 3 has $E_{\mathrm{full}} = 0$ and $B_{\mathrm{ring}} = 2(2-\sqrt{2})\sqrt{5} \approx 2.6197 > 0$ and $B^{xy}_{\mathrm{3D}} \approx 1.5279 > 0$.*

(b) For every arrangement of every order, $V(S) \geq \sum_c z_c^2 / \max_c |v_c| > 0$; consequently the three-dimensional vector sum $\sum_c z_c u'_c$ is nonzero for every *arrangement, magic or not.*

(c) *If $u_c = (F(x_c), G(y_c))$ with functions F, G of one coordinate each (for example the linearisation $u_c = (x_c, y_c)/r_0$ with a constant r_0), then $E_{\mathrm{full}}(S) = 0$ implies $\sum_c z_c u_c = 0$ (Theorem 4.1); the radial unit vectors do not have this form.*

Proof. (a) By D_4-invariance it suffices to take the Lo Shu square with rows $(2,7,6), (9,5,1), (4,3,8)$, so $z = s - 5$. The four edge cells have $u = (\pm 1, 0), (0, \pm 1)$ and heights $z = -4$ (right), 4 (left), 2 (top), -2 (bottom), contributing $(-8, 4)$; the four corners have $u = (\pm 1, \pm 1)/\sqrt{2}$ and heights $-3, 1, -1, 3$ (upper left, upper right, lower left, lower right), contributing $(8, -4)/\sqrt{2} = (4\sqrt{2}, -2\sqrt{2})$. The total is $(4\sqrt{2} - 8,\ 4 - 2\sqrt{2}) = (4 - 2\sqrt{2})(-2, 1)$, of norm $(4 - 2\sqrt{2})\sqrt{5}$. The value of $B_{3\mathrm{D}}^{xy}$ is obtained in the same way, numerically. (b) $V = \sum_c z_c \cdot z_c/|v_c|$ and $z_c \neq 0$ for all but at most one cell. (c) is Theorem 4.1(b). □

Remark 4.4 (Reading of Proposition 4.3). The negative statement is about implication only. Statement (a) shows that $E_{\mathrm{full}} = 0$ does not imply the vanishing of the ring imbalance, and (b) shows that the vanishing of a full three-dimensional unit-vector sum is impossible for *any* arrangement, because the vertical component is a sum of squares. The comparison below shows, however, that magic squares are not typical in the unit-vector imbalance either.

Numerical comparison, $n = 4$. Table 2 compares the 880 essentially different magic squares of order 4 with 3×10^5 uniformly random arrangements (fixed seed, `anc/balance_check.py`). The linear imbalance vanishes on every magic square, as it must (Theorem 4.1), and is nonzero for 99.97% of random arrangements. The unit-vector imbalances of magic squares are smaller than those of random arrangements on average, the mean over magic squares lying below the 7.4% (ring) and 2.4% (three-dimensional, horizontal) quantiles of the random distribution; but they never vanish (smallest values 0.070 and 0.036), and the largest magic value of B_{ring} is the 31st percentile of the random distribution (Figure 2(b,c)). Among random arrangements the Spearman correlation between E_{full} and B_{lin}, B_{ring}, $B_{3\mathrm{D}}^{xy}$ is 0.38, 0.33, 0.34. The vertical component V of the three-dimensional model is 59.57 on average for magic and for random arrangements alike: it is close to $\sum_c |z_c| = 64$, a constant, because $|v_c| \approx |z_c|$ for most cells, and carries no information about the line sums. For $n = 3$, over all 9! arrangements the smallest value of B_{ring} is 0.131, and 9.1% of the arrangements have a smaller ring imbalance than the magic squares (2.62).

	880 magic squares ($n = 4$)			3×10^5 random	
imbalance	mean	min	max	mean	rank of magic mean
$B_{\mathrm{lin}} = \lvert\sum_c z_c(x_c, y_c)\rvert$	0	0	0	27.03	0.03%
B_{ring}	5.51	0.070	11.98	17.06	7.4%
$B_{3\mathrm{D}}^{xy}$	0.874	0.036	2.54	4.90	2.4%
$\lvert\sum_c z_c x_c y_c\rvert$	19.9	0	60	19.3	57%

Table 2: Balance functionals on the 880 essentially different magic squares of order 4 and on 3×10^5 uniformly random arrangements. "Rank" is the percentage of random arrangements with a smaller value than the mean over the magic squares. The mixed moment $\lvert\sum_c z_c x_c y_c\rvert$ is not controlled by E_{full} and is indistinguishable between the two groups.

Remark 4.5 (A relaxed gradient flow). Let z be a real-valued height function and consider the gradient flow $\dot{z} = -\nabla E_{\mathrm{full}}(z)$ of the quadratic form (5). It is linear, and its limit is the orthogonal projection of the initial height onto the zero set $W^{\perp}$, i.e. $z_\infty = z_0 - P_W z_0$. By Theorem 4.1 the limit has $E_{\mathrm{full}} = 0$ and exactly zero linear imbalance. It does not balance the ring: starting from uniformly random permutations for $n = 4$ (2×10^4 starts), the mean of B_{ring} decreases from 17.1 to 3.9 (and that of $B_{3\mathrm{D}}^{xy}$ from 4.9 to 1.5), so the limit is balanced linearly but not on the ring.

4.3 Multipole moments

Remark 4.6 (Multipole moments). The sums $m_{ab} = \sum_c z_c x_c^a y_c^b$ are the Cartesian moments of the signed planar distribution z; the monopole is m_{00}, the dipole (m_{10}, m_{01}), and the quadrupole is

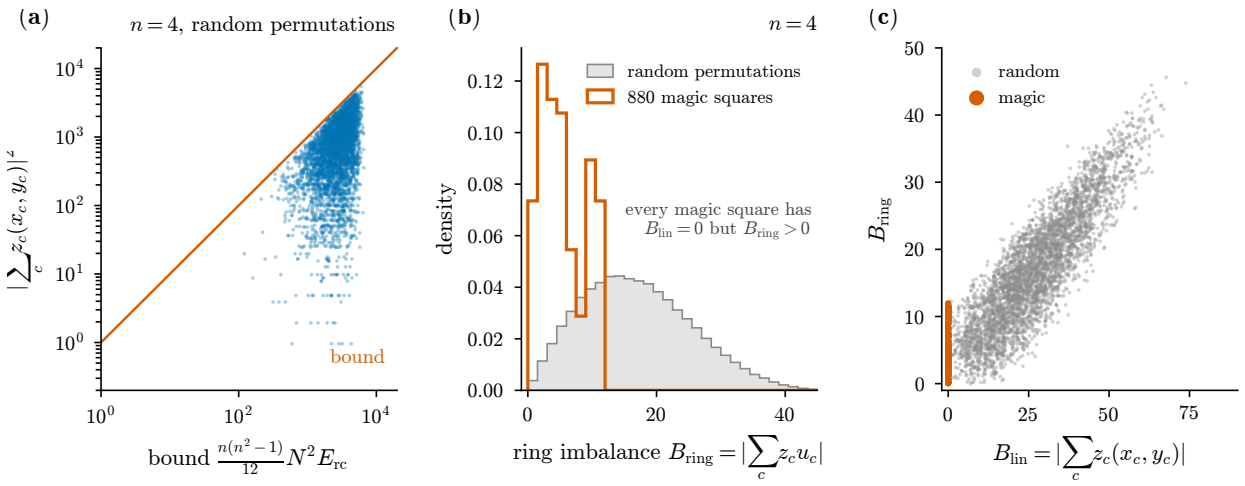


Figure 2: Balance of the signed mass for $n = 4$. (a) The bound (15): the squared linear imbalance $|\sum_c z_c(x_c, y_c)|^2$ against $\frac{n(n^2-1)}{12}N^2E_{\mathrm{rc}}$ for 4000 random arrangements; the line is the bound, which is attained. (b) Distribution of the ring imbalance $B_{\mathrm{ring}} = |\sum_c z_c u_c|$ for 3×10^5 random arrangements (grey) and for the 880 magic squares (orange outline): magic squares are smaller than typical but never zero. (c) Linear against ring imbalance: every magic square lies on the axis $B_{\mathrm{lin}} = 0$ at a positive value of B_{ring}.

built from m_{20}, m_{02}, m_{11}. Rogers and Loly studied the electric multipole expansion of a magic cube regarded as a charge distribution [19]. In the present notation, Theorem 4.1 says that for a magic square every *pure* moment m_{a0}, m_{0b} vanishes, so the monopole, the dipole and the diagonal quadrupole components m_{20}, m_{02} (hence the trace and the traceless part $m_{20}-m_{02}$) vanish, whereas the mixed moments m_{ab} with $a, b \geq 1$, beginning with m_{11}, are not determined by the line sums. The hierarchy of Proposition 3.11 is the statement that, for real height functions, the vanishing of the pure axial moments up to degree $n-1$ in each coordinate is *equivalent* to equal row and column sums, and that fewer moments do not suffice. We make no claim beyond this algebraic correspondence.

5 Computations

This section reports the computations behind the statements of Section 3 and the exact count of the low-mode class for $n = 4$. All counts and all swap gaps were computed in exact integer or rational arithmetic; Monte Carlo results are reported with standard errors. Hardware, software and algorithms are described in Section A; every number below is produced by a script in the ancillary directory `anc/` (see its `README`).

5.1 Exhaustive enumeration for $n = 3$

All $9! = 362{,}880$ arrangements of order 3 were enumerated in exact integer arithmetic. Exactly 8 arrangements have $E_{\mathrm{full}} = 0$ (the Lo Shu orbit), and the same 8 are the zeros of E_{low} (Remark 3.17); 760 arrangements satisfy the degree-one conditions, 4,224 have both diagonals summing to 15, and 72 are semi-magic. Both energies have mean 40/27 (the coincidence is special to $n = 3$: $\mathrm{Var}(X) = \frac{n^2-1}{12}$ equals $n\mathrm{Var}(C_j) = \frac{n-1}{n}$ only there), with standard deviations 0.4266 (E_{full}) and 0.8370 (E_{low}), modes 138/81 and 90/81, maxima 180/81 and 360/81, and 177 and 346 distinct values, all multiples of 1/81. Under single swaps the symmetric E_{full} has 16 local minima, the 8 magic squares and 8 non-magic arrangements at $E_{\mathrm{full}} = 2/81$ (two line deviations of ± 1); E_{low} is more rugged, with 344 local minima, 336 of them non-magic.

5.2 The energy of a random arrangement

Let S be a uniformly random arrangement, i.e. the values $1,\dots,N$ are placed in the grid by a uniformly random permutation π. For a test function φ on the grid write $T_\varphi := \mathrm{Cov}(\varphi, Z) = \sum_c a_c\, z_{\pi(c)}$ with $a_c = (\varphi_c - \bar\varphi)/N$; this is a linear permutation statistic in the sense of Section 2.3, and its first two moments follow from (2). Throughout, $\mathrm{Var}(\varphi)$ and $\mathrm{Cov}(\varphi,\psi)$ denote population variances and covariances over the N cells, and

$$\sigma_n^2 := \frac{\mathrm{Var}(Z)}{N-1} = \frac{N+1}{12} = \frac{n^2+1}{12}, \qquad \mathrm{Var}(Z) = \frac{N^2-1}{12}.$$

Proposition 5.1 (Exact moments of the random energies)**.** *For a uniformly random arrangement and any test functions φ,ψ,*

$$\mathbb{E}[T_\varphi] = 0, \qquad \mathbb{E}[T_\varphi T_\psi] = \sigma_n^2\,\mathrm{Cov}(\varphi,\psi). \tag{16}$$

Consequently, with E_{low} as in (6) *and the symmetric E_{full} (all n rows, all n columns, both diagonals; $2n+2$ terms),*

$$\mathbb{E}\,\mathrm{Cov}(X,Z)^2 = \mathbb{E}\,\mathrm{Cov}(Y,Z)^2 = \frac{n^4-1}{144}, \qquad \mathbb{E}\,\mathrm{Cov}(D,Z)^2 = \mathbb{E}\,\mathrm{Cov}(R_k,Z)^2 = \frac{(n-1)(n^2+1)}{12\,n^2}, \tag{17}$$

$$\mathbb{E}[E_{\mathrm{low}}] = \frac{(n^2+1)\,(n^4-n^2+12n-12)}{72\,n^2} = \frac{n^4}{72} + \frac{n}{6} - \frac{13}{72} + \frac{1}{6n} - \frac{1}{6n^2}, \tag{18}$$

$$\mathbb{E}[E_{\mathrm{full}}] = \frac{n^4-1}{6\,n^2} = \frac{n^2}{6} - \frac{1}{6n^2}. \tag{19}$$

The covariance matrix $\Sigma_n := \sigma_n^2 G$ of the vector of covariances is σ_n^2 times the Gram matrix G of the centred test functions. For E_{low}, $G = \mathrm{diag}\big(\frac{n^2-1}{12}, \frac{n^2-1}{12}, \frac{n-1}{n^2}, \frac{n-1}{n^2}\big)$ except that the two diagonal indicators have $\mathrm{Cov}(D_{\mathrm{m}}, D_{\mathrm{a}}) = -1/n^2$ when n is even (they are uncorrelated when n is odd). For E_{full}, G is block diagonal: the n row indicators have Gram block $\frac{1}{n}(I - \frac{1}{n}J)$, the columns likewise, every row–column, row–diagonal and column–diagonal covariance vanishes, and the diagonal block is as for E_{low}. Hence the eigenvalues of Σ_n are

$$E_{\mathrm{low}}:\ \sigma_n^2\Big\{\tfrac{n^2-1}{12}\,(\times 2),\ \tfrac{n-1}{n^2}\,(\times 2)\Big\}, \qquad E_{\mathrm{full}}:\ \sigma_n^2\Big\{\tfrac{1}{n}\,(\times\, 2n-2),\ \tfrac{n-1}{n^2}\,(\times 2)\Big\} \tag{20}$$

for odd n; for even n the last pair $\{\frac{n-1}{n^2}, \frac{n-1}{n^2}\}$ is replaced by $\{\frac{1}{n}, \frac{n-2}{n^2}\}$. Finally, the variances are explicit rational functions of n, given as (22) *in Section B, with $\mathrm{Var}(E_{\mathrm{low}}) = n^8/5184 + O(n^7)$ and $\mathrm{Var}(E_{\mathrm{full}}) = n^3/36 + O(n^2)$.*

Proof. Since $\sum_c a_c = 0$, $\mathbb{E}[T_\varphi] = \bar z \sum_c a_c = 0$. For two linear permutation statistics $T_a = \sum_i a_i b_{\pi(i)}$, $T_{a'} = \sum_i a'_i b_{\pi(i)}$ the classical identity (2) reads $\mathrm{Cov}(T_a, T_{a'}) = \frac{1}{N-1}\sum_i (a_i - \bar a)(a'_i - \bar a')\sum_j (b_j - \bar b)^2$. With $a = (\varphi - \bar\varphi)/N$, $a' = (\psi - \bar\psi)/N$, $b = z$ this is $\frac{1}{N-1}\cdot\frac{N\mathrm{Cov}(\varphi,\psi)}{N^2}\cdot N\mathrm{Var}(Z) = \sigma_n^2\mathrm{Cov}(\varphi,\psi)$, which is (16). The entries of G are elementary: $\mathrm{Var}(X) = \mathrm{Var}(Y) = \frac{n^2-1}{12}$; a line indicator has $\mathrm{Var}(L) = \frac{1}{n}(1-\frac{1}{n})$; two lines meeting in m cells have $\mathrm{Cov}(L,L') = \frac{m}{N} - \frac{1}{n^2}$ (Lemma 3.6); and $\mathrm{Cov}(X,Y) = 0$, $\mathrm{Cov}(X,D) = \mathrm{Cov}(Y,D) = 0$ since a diagonal contains every column and every row coordinate exactly once. Taking traces of Σ_n gives (17)–(19): $\mathbb{E}[E_{\mathrm{low}}] = \sigma_n^2\big(2\cdot\frac{n^2-1}{12} + 2\cdot\frac{n-1}{n^2}\big)$ and $\mathbb{E}[E_{\mathrm{full}}] = \sigma_n^2(2n+2)\frac{n-1}{n^2}$. The eigenvalues (20) follow from the block structure, $\frac{1}{n}(I - \frac{1}{n}J)$ having eigenvalue $\frac{1}{n}$ with multiplicity $n-1$ and 0 once.

The variances are obtained by expanding $\mathbb{E}[E^2]$ over the equality patterns of cell quadruples and using the exact fourth moments of a uniform permutation (Section A); the result is a rational number for each n, and the closed forms (22) were obtained by exact rational interpolation on $n = 3,\dots,45$, verified on 5–11 further orders each and against the exhaustive $n = 3$ values $\mathrm{Var}(E_{\mathrm{low}}) = 1532/2187$, $\mathrm{Var}(E_{\mathrm{full}}) = 398/2187$. For $n = 3,\dots,6$ the exact means and variances were also recomputed by a separate direct summation over the equality patterns of cell quadruples (`anc/verify_numbers.py`) and agree with the closed forms. □

Corollary 5.2 (Limiting law)**.** *As $n \to \infty$, the standardised vector*

$$\big(T_X/\sqrt{\Sigma_{11}},\ T_Y/\sqrt{\Sigma_{22}},\ T_{D_\mathrm{m}}/\sqrt{\Sigma_{33}},\ T_{D_\mathrm{a}}/\sqrt{\Sigma_{44}}\big)$$

converges in distribution to a standard Gaussian vector in $\mathbb{R}^4$, and therefore

$$\frac{E_{\mathrm{low}}}{(n^4-1)/144} \xrightarrow{\ d\ } \chi^2_2, \qquad \textit{equivalently} \qquad \frac{E_{\mathrm{low}}}{\mathbb{E}[E_{\mathrm{low}}]} \xrightarrow{\ d\ } \mathrm{Exp}(1). \tag{21}$$

For finite n the Gaussian approximation gives $E \approx \sum_j w_j \chi^2_{1,j}$ with independent χ^2_1 variables and weights w_j the eigenvalues (20)*; its density is computed by numerical inversion of the characteristic function (Imhof's method [10]) in Figure 3.*

Proof. Each fixed linear combination $\sum_k \alpha_k T_{\varphi_k}$ is again a linear permutation statistic $\sum_c a_c z_{\pi(c)}$. Hoeffding's theorem [9, Thm. 3] (Section 2.3) asserts asymptotic normality provided that, for every $r \geq 3$, $N^{-1}\sum_{i,j}|d_N(i,j)|^r$ is $O\big((N^{-1}\sum_{i,j} d_N(i,j)^2)^{r/2}\big)$, d_N the doubly centred array. In the product case $c_N(i,j) = a_i z_j$ the ratio of the two sides equals $N^{1-r/2}\kappa_r(a)\kappa_r(z)$ with κ_r the standardised r-th absolute moments. For z (uniform on $\{1,\dots,N\}$, centred) $\kappa_r(z) = O(1)$; for each of X, Y and the two diagonal indicators, $\kappa_r(a) = O(n^{r/2-1}) = O(N^{r/4-1/2})$ (the worst case is an indicator of n of the N cells, a Bernoulli$(1/n)$ variable), so the ratio is $O(N^{1/2-r/4}) \to 0$. The same bound holds for every fixed combination $a = \sum_k \alpha_k a_k$ of the four standardised test functions: $\|a\|_r \leq \sum_k |\alpha_k| \|a_k\|_r$ by Minkowski's inequality, while $\mathrm{Var}(\sum_k \alpha_k T_k) \geq (1-o(1))\sum_k \alpha_k^2$ because the only nonzero correlation among the T_k tends to zero. By the Cramér–Wold device the vector is asymptotically Gaussian with covariance Σ_n; for even n the only nonzero correlation is $\mathrm{corr}(T_{D_\mathrm{m}}, T_{D_\mathrm{a}}) = -1/(n-1) \to 0$, so the standardised limit is $N(0, I_4)$. In $E_{\mathrm{low}} = \Sigma_{11}(\tilde T_X^2 + \tilde T_Y^2) + \Sigma_{33}\tilde T_{D_m}^2 + \Sigma_{44}\tilde T_{D_a}^2$ the diagonal terms carry weight $\Sigma_{33}/\Sigma_{11} = 12(n-1)/(n^2(n^2-1)) \to 0$, giving (21); the second form uses $\mathbb{E}[E_{\mathrm{low}}]/((n^4-1)/72) \to 1$. □

Remark 5.3. For E_{full} the number of statistics grows with n, so the corollary does not apply verbatim; the Gaussian mixture $\sigma_n^2\big[\frac{1}{n}\chi^2_{2n-2} + \frac{n-1}{n^2}\chi^2_2\big]$ is only a heuristic at the orders we can simulate, and the ratio of the exact to the Gaussian-mixture variance is $0.24, 0.45, 0.56, 0.64, 0.69, 0.73$ for $n = 3,\dots,8$ (for E_{low}: $0.51, 0.76, 0.86, 0.90, 0.93, 0.95$). For $n = 3$, E_{low} takes 346 and E_{full} 177 distinct values, all multiples of $1/81$; the Kolmogorov distances 0.10 and 0.20 at $n = 3$ are inflated by this discreteness and are not a failure of the Gaussian approximation as such.

n	energy	$\mathbb{E}[E]$ exact	MC mean	$\sqrt{\mathrm{Var}E}$ exact	MC std	Gauss. std	$\lvert\delta_{\mathrm{mean}}\rvert$	KS
3	E_{low}	$40/27 = 1.4815$	1.4814	0.8370	0.8369	1.171	6×10^{-5}	0.103
3	E_{full}	$40/27 = 1.4815$	1.4815	0.4266	0.4266	0.869	8×10^{-6}	0.200
4	E_{low}	$391/96 = 4.0729$	4.0736	3.1284	3.1288	3.586	2×10^{-4}	0.037
4	E_{full}	$85/32 = 2.6562$	2.6564	0.9074	0.9073	1.349	7×10^{-5}	0.107
5	E_{low}	$234/25 = 9.3600$	9.3605	8.0483	8.0482	8.694	6×10^{-5}	0.019
5	E_{full}	$104/25 = 4.1600$	4.1600	1.3966	1.3966	1.867	9×10^{-6}	0.078
6	E_{low}	$2035/108 = 18.843$	18.843	17.122	17.122	18.007	4×10^{-5}	0.012
6	E_{full}	$1295/216 = 5.9954$	5.9957	1.9671	1.9671	2.459	6×10^{-5}	0.061
7	E_{low}	$5050/147 = 34.354$	34.349	32.174	32.167	33.349	1×10^{-4}	0.008
7	E_{full}	$400/49 = 8.1633$	8.1631	2.5697	2.5698	3.089	2×10^{-5}	0.050
8	E_{low}	$22295/384 = 58.060$	58.058	55.377	55.381	56.888	3×10^{-5}	0.007
8	E_{full}	$1365/128 = 10.664$	10.664	3.2352	3.2353	3.777	3×10^{-5}	0.042

Table 3: Exact moments of Proposition 5.1 against Monte Carlo (10^8 uniformly random arrangements per order; for $n = 3$ the exact values were also confirmed by enumerating all 9! arrangements). $|\delta_{\mathrm{mean}}|$ is the relative error of the MC mean; all MC means and variances agree with the exact values to within 5×10^{-4} relative error, consistent with the sampling standard errors. "Gauss. std" is the standard deviation of the Gaussian-limit mixture $\sum_j w_j\chi^2_1$ of Corollary 5.2, and KS is the Kolmogorov distance between the MC sample and that mixture (not the Exp(1) limit). Monte Carlo data: `anc/mc_results.json`.

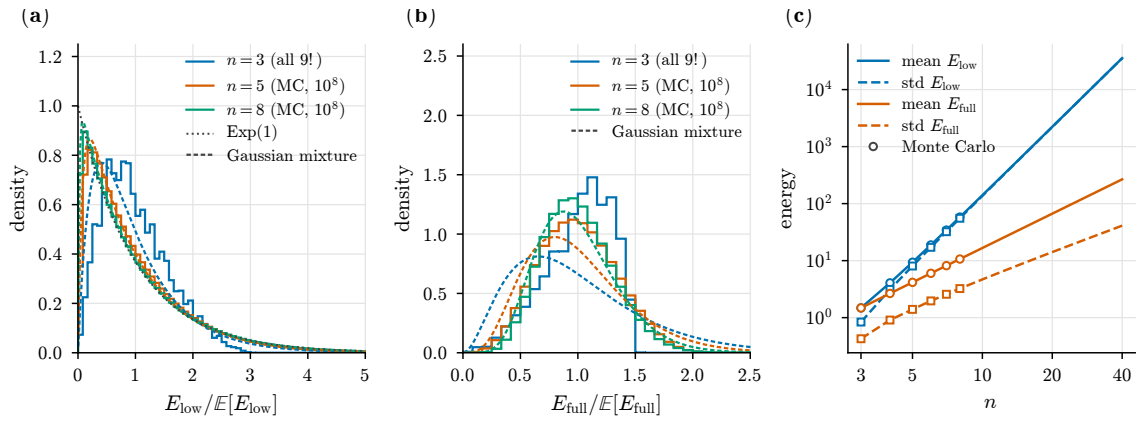


Figure 3: Energy of a uniformly random arrangement. (a) Distribution of $E_{\mathrm{low}}/\mathbb{E}[E_{\mathrm{low}}]$ for $n = 3$ (exact, all 9! arrangements) and $n = 5, 8$ (10^8 Monte Carlo samples each), with the Gaussian-limit mixture densities (dashed, characteristic-function inversion with the weights (20)) and the $n \to \infty$ limit Exp(1) of Corollary 5.2 (dotted). (b) The same for the symmetric E_{full}; here the Gaussian mixture is only a slowly improving approximation (see the remark after Corollary 5.2). (c) Exact mean (solid) and standard deviation (dashed) of both energies from Proposition 5.1, $3 \le n \le 40$, with the Monte Carlo values for $n = 3, \dots, 8$ (open markers). The mean of E_{low} grows like $n^4/72$ and its standard deviation like $n^4/72$ as well (an exponential limit has unit coefficient of variation); E_{full} has mean $\sim n^2/6$ and standard deviation $\sim n^{3/2}/6$.

5.3 Swap gaps: exact verification

Theorem 3.18 gives, for a magic square S and the swap of two cells a, b with $d = |s_a - s_b|$, $E_{\mathrm{full}}(S^{ab}) = d^2 m(a,b)/N^2$ with $m(a,b) \in \{2, \dots, 6\}$ the number of lines separating a from b. We verified the formula in exact rational arithmetic, with the symmetric energy evaluated directly from the definition, on every one of the $880 \cdot \binom{16}{2} = 105{,}600$ single swaps of the 880 essentially different order-4 squares, the $8 \cdot \binom{9}{2} = 288$ swaps of the eight order-3 squares, and all swaps of the Siamese (de la Loubère) squares of orders 5, 7 and 9: agreement in every case, and $E_{\mathrm{full}}(S) = 0$ exactly for every magic square. The gap spectrum is arithmetic, not "banded": over the 105,600 order-4 swaps $N^2 \Delta E = d^2 m$ takes exactly 70 distinct integer values from 2 to $980 = 14^2 \cdot 5$, and the minimum gap $N^2 \Delta_{\min}(S)$ is 2 for 432 squares, 3 for 420 and 4 for 28 (Corollary 3.20(3)); it is 3 for all eight order-3 squares and 3, 2, 2 for the Siamese squares of orders 5, 7, 9. Figure 4 shows the structure: the values $d^2 m$ arise from the 15×5 pairs (d, m), so the spectrum is a union of the scaled square sequences $m, 4m, 9m, \dots$, $2 \le m \le 6$.

5.4 The low-mode balanced class for $n = 4$

Write L_n for the set of arrangements of $\{1, \dots, N\}$ with $\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = \mathrm{Cov}(D_{\mathrm{m}}, Z) = \mathrm{Cov}(D_{\mathrm{a}}, Z) = 0$, i.e. $E_{\mathrm{low}} = 0$ (Corollary 3.12). Each of the four conditions is a *linear* statistic of the arrangement: with cells indexed by $(i, j) \in \{0, \dots, n-1\}^2$, $\mathrm{Cov}(X, Z) = 0 \iff T_1 := \sum_{i,j} j\, s_{ij} = \frac{n-1}{2} \cdot \frac{N(N+1)}{2}$, $\mathrm{Cov}(Y, Z) = 0 \iff T_2 := \sum_{i,j} i\, s_{ij}$ equals the same target, and the two diagonal conditions say that $T_3 = \sum_i s_{ii}$ and $T_4 = \sum_i s_{i,n-1-i}$ equal $M(n) = n(N+1)/2$.

Proposition 5.4 (Exact size of L_4)**.** $|L_3| = 8$ *(the Lo Shu orbit; the class with only* $\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = 0$ *has* 760 *members), and*

$$|L_4| \;=\; 14\,639\,440 \;=\; 8 \times 1\,829\,930,$$

of which exactly 7040 *are magic squares. The class is* D_4*-invariant and* D_4 *acts freely on it (*1 829 930 *orbits of size* 8*). The first-moment class* $\{\mathrm{Cov}(X, Z) = \mathrm{Cov}(Y, Z) = 0\}$ *has* 6 697 177 424 *members for* $n = 4$*, and the class with both diagonals equal to* 34 *has* 49 932 288 000 *members.*

Computation. The count is a meet-in-the-middle enumeration over the four additive statistics $T_1, \dots, T_4$ (Section A; 9 s on 16 threads), with the matching pairs expanded explicitly to obtain

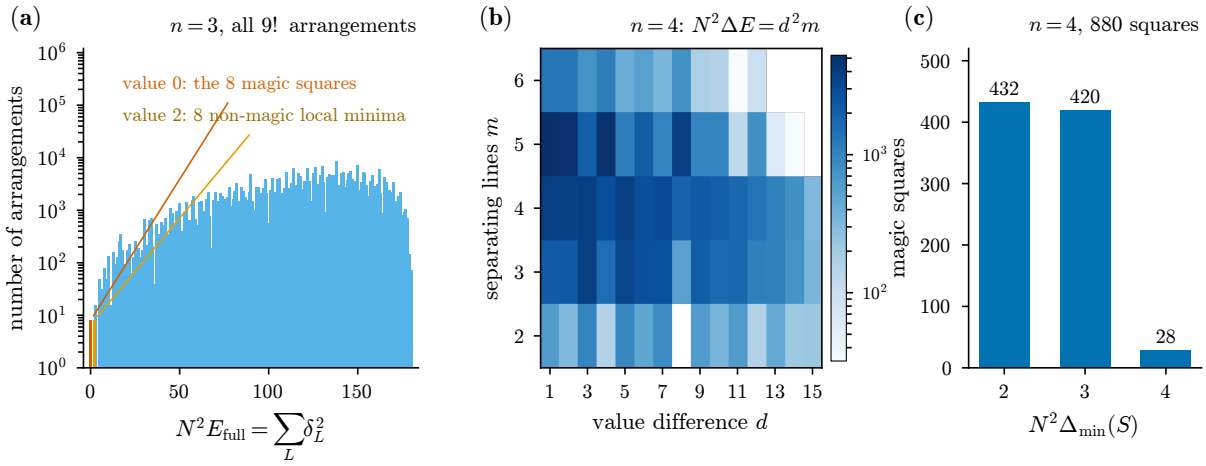


Figure 4: Energy landscape and swap gaps. (a) For $n = 3$, the number of arrangements (of all 9!) with each value of $N^2E_{\text{full}} = \sum_L \delta_L^2$: the 8 magic squares at value 0, and the 8 non-magic local minima, which lie at value 2. (b) For $n = 4$, the number of single swaps of the 880 essentially different magic squares (105,600 in total) by value difference d and number m of separating lines; the energy gap is exactly $N^2\Delta E = d^2 m$, which takes 70 distinct values between 2 and 980. (c) The smallest gap $N^2\Delta_{\min}(S)$ over the 880 squares.

the row- and column-sum breakdown of Table 4. Validation: on $n = 3$ the same program reproduces the brute-force counts 8, 760, 4224, 8 magic and 72 semi-magic; on $n = 4$, with the $2n$ line sums in place of T_1, T_2, it reproduces the 7040 magic and the $549{,}504 = 8 \times 68{,}688$ semi-magic squares; the diagonal-only count was also obtained by direct combinatorics (ordered pairs of disjoint 4-subsets of $\{1, \ldots, 16\}$ with sum 34: 2150 pairs, times $(4!)^2\, 8!$), and the probability $|L_4|/16!$ was checked by 2×10^9 Monte Carlo samples (1373 hits against 1399 expected; relative standard error 2.7%). □

Table 4: Breakdown of L_4 by the number of row sums and column sums equal to $M(4) = 34$. Exactly two or three equal row sums are impossible: if $R_i = 34 + e_i$ then $\sum_i e_i = 0$ and $\sum_i i\, e_i = 0$ (this is T_2), which cannot hold with exactly one or two nonzero e_i; the same holds for columns. The table is symmetric under transposition.

	columns equal to 34			
rows equal to 34	0	1	4	total
0	9 637 312	2 172 824	100 728	11 910 864
1	2 172 824	429 312	9 336	2 611 472
4	100 728	9 336	**7 040** (magic)	117 104
total	11 910 864	2 611 472	117 104	14 639 440

Thus only $7040/14\,639\,440 = 0.048\%$ of the low-mode balanced arrangements are magic; 110 064 members have all row sums equal to 34 but not all column sums (and symmetrically), and 9 637 312 (66%) have *no* row or column summing to 34 at all; the arrangement of Remark 3.3 belongs to the last class.

Frequency under random sampling. $|L_4|/16! = 6.997 \times 10^{-7}$, about one arrangement in 1.43 million, so naive sampling rarely sees a member of L_4. Among the 10^8 uniformly random arrangements of Section 5.2, 61 lie in L_4 (expected 70.0 ± 8.4), and 2×10^9 further samples (sixteen independent streams) gave 1373 members, against an expectation of 1399. A Gaussian local-limit estimate explains the order of magnitude of $|L_4|$: under a uniformly random arrangement the vector $(T_1, \ldots, T_4)$ has mean equal to the target vector (by symmetry) and covariance Σ given by (2) (T_1, T_2 are uncorrelated with each other and with the diagonals; the two diagonal sums have correlation $-1/(n-1)$ for even n and 0 for odd n), and the $\mathbb{Z}$-span of the change vectors of all

single transpositions has index 1 in $\mathbb{Z}^4$ (Smith normal form, computed for $n \le 8$), so

$$|L_n| \;\approx\; N!\,(2\pi)^{-2}\det(\Sigma)^{-1/2}.$$

The estimate gives 13.6 for $n = 3$ (true: 8, ratio 0.59) and 1.82×10^7 for $n = 4$ (true: 14 639 440, ratio 0.80); for the two-condition classes the ratios at $n = 3, 4$ are $0.59, 0.91$ (first moments) and $1.10, 0.96$ (diagonals). For $n = 5, 6$ we have no exact count; the estimate gives 6.7×10^{17} and 1.5×10^{33}, and Monte Carlo (10^{10} and 2×10^{10} samples, 328 and 62 hits) gives 5.1×10^{17} ($\pm 6\%$) and 1.2×10^{33} ($\pm 13\%$), i.e. ratios 0.76 and 0.78 to the estimate; these are estimates of the order of magnitude only. For $n = 3$ the classes coincide (L_3 is the set of magic squares), while for $n = 4$ the magic squares are 0.048% of L_4. Heuristically this fraction tends to zero: the four low-mode conditions cost a factor of order $\det(\Sigma)^{1/2} \propto n^{13}$, while a magic square requires $2n$ independent line conditions, each costing a further factor of order $n^{5/2}$.

6 Discussion

6.1 What the orthogonality viewpoint buys, and what it does not

Reading the line-sum conditions as $z \perp W$ changes nothing about which arrangements are magic, but it changes which questions are natural, and four of them have been answered above. First, the test space can be varied: replacing the $2n$ line indicators by the polynomials x^m, y^m ($m \le k$) gives the moment hierarchy, Proposition 3.11 locates the degree $k = n - 1$ at which polynomial test functions become equivalent to line sums for real heights, and Theorem 3.15 shows that the permutation constraint is not idle: for $n \ge 14$ integrality forces equal line sums one degree earlier than linear algebra does. In this frame the four-term energy E_{low} is the degree-one member and is weaker than "magic" for every $n \ge 4$, while Loly's invariance of the moment of inertia [11] is a consequence of the degree-two member: the value $n^2(n^4 - 1)/12$ is the constant term of (10), the arrangement-dependent part is the single cross-moment $\mathrm{Cov}(Z, X^2 + Y^2)$, and the invariance is weaker than the degree-two level (Corollary 3.13). Second, $z \perp W$ is a statement about the signed mass z: all separable moments vanish (Theorem 4.1), the centre-of-mass offset of an arbitrary arrangement is controlled by E_{full} with a sharp constant (Theorem 4.2), and the line-sum conditions say nothing about mixed moments or about unit-vector sums (Proposition 4.3). Third, each $\mathrm{Cov}(\varphi, Z)$ is a linear permutation statistic, so the energies of a random arrangement have exact moments and a Hoeffding limit law (Proposition 5.1, Corollary 5.2), and Loly's moment of inertia has an exact null distribution (Corollary 3.14); the growth $\mathbb{E}[E_{\text{low}}] \sim n^4/72$, $\mathbb{E}[E_{\text{full}}] \sim n^2/6$ is a property of the normalisation $\mathrm{Cov} = \frac{1}{N}\sum$, not of the arrangements. Fourth, because $N^2 E_{\text{full}}$ is a sum of squared integer deviations, a swap changes it by the two-line formula (12), which gives local minimality and the exact gap spectrum for all n at once (Corollary 3.20).

What the viewpoint does not buy is equally clear. The projection lemma is a restatement of the definition (Remark 3.9); it proves nothing about magic squares that the line sums do not already say, and we have not used it to count anything: the counting is done by the meet-in-the-middle enumeration of Proposition 5.4. The low-mode energy is a poor proxy for the magic property: for $n = 4$ its zero set is about 2080 times larger than the set of magic squares, and even its two-condition sub-classes have 6.7×10^9 and 5.0×10^{10} members.

6.2 Local search on the energy

By Theorem 3.18 a single swap changes E_{full} by an explicit quadratic expression in the swapped values, so local search is cheap. We ran two elementary searches on the symmetric E_{full} from uniformly random starting arrangements (Table 5): steepest swap descent (take the swap with the largest decrease; stop when none decreases the energy) and a plain simulated annealing (random swap proposals, Metropolis acceptance, geometric cooling from $T_0 =$ the median uphill increment of $N^2 E_{\text{full}}$ at a random arrangement to $T_1 = 1/2$, over $2 \times 10^4 N$ proposals). Descent reaches a

magic square from 19.5% of random starts for $n = 4$ but from only 1.2% ($n = 5$) and 0.8% ($n = 6$); otherwise it stops after 8–14 swaps on average in a non-magic local minimum with mean residual $N^2 E_{\text{full}} \approx 4.4$ (smallest observed residual 1 for $n = 5, 6$ and 2 for $n = 4$). Annealing succeeds in every run for $n = 4$ and in 78% ($n = 5$) and 50% ($n = 6$) of runs within the budget; the failed runs end with a best residual $N^2 E_{\text{full}}$ of 1.05 ($n = 5$) and 1.2 ($n = 6$) on average. The landscape is thus easy to descend but, as the exhaustive $n = 3$ count of non-magic local minima already suggests (eight, all at $E_{\text{full}} = 2/81$), it contains many shallow non-magic minima that a swap move cannot leave. The experiment is small and the methods elementary; neither competes with the classical constructions or with the backtracking enumerations of [3, 21]. The energy is a search objective, not a counting tool.

	steepest swap descent (2×10^5 runs)			simulated annealing (400 runs)		
n	success	mean swaps	stuck ($E_{\text{full}} > 0$)	success	median proposals	accepted swaps
4	0.195 ± 0.001	8.1	80.5%	1.000	1.5×10^5	4.2×10^4
5	0.0115 ± 0.0002	11.2	98.9%	0.78 ± 0.02	4.2×10^5	8.0×10^4
6	0.0079 ± 0.0002	14.1	99.2%	0.50 ± 0.03	6.6×10^5	1.1×10^5

Table 5: Local search on the symmetric E_{full} from uniformly random arrangements. "Success" is the fraction of runs reaching $E_{\text{full}} = 0$ (a magic square), $\pm$ one standard error; "stuck" is the fraction of descents ending in a non-magic local minimum; "mean swaps" is over all descents. Annealing budget $2 \times 10^4 N$ proposals; "median proposals" and "accepted swaps" (mean) refer to successful runs. Full-scale data: `anc/algo_results.json`; an independent rerun at reduced scale (2×10^4 descents, 64 annealing runs per order, `anc/search_reduced.json`) gives success rates 0.193, 0.011, 0.008 for descent and 1.00, 0.81, 0.45 for annealing, within sampling error of the table.

6.3 Open questions

1. *Size of the low-mode class.* $|L_3| = 8$, $|L_4| = 14{,}639{,}440$, and the Gaussian local-limit estimate $N!(2\pi)^{-2} \det(\Sigma)^{-1/2}$ is within 20% at $n = 4$ and, by Monte Carlo, within about 25% at $n = 5, 6$ (Section 5.4). Is $|L_n|/N! \sim c\, n^{-13}$ with an explicit constant, i.e. does a local limit theorem hold for the four lattice statistics $(T_1, \dots, T_4)$?
2. *The permutation threshold.* What is $k^*(n)$ for $n \geq 14$? The proof of Theorem 3.15 compares the entries of the integer vectors orthogonal to all polynomials of degree $\leq k$ with the largest possible deviation $n^2(n-1)/2$. If the smallest such vectors were always alternating binomial vectors supported on $k+2$ consecutive columns, whose largest entry is $\binom{k+1}{\lfloor (k+1)/2 \rfloor} \approx 2^{k+1}/\sqrt{k}$, the comparison $2^k \gtrsim n^3$ would give $k^*(n) = O(\log n)$; but shorter integer vectors with small entries might exist, and we have no proof.
3. *Exact unit-vector balance.* Is there a magic square of any order with $B_{\text{ring}} = 0$? None exists for $n = 3$ and none among the 880 classes for $n = 4$ (smallest value 0.070); the orthogonality conditions give no handle on this because the radial unit vectors are not in the line space (Proposition 4.3).
4. *Other square families.* The orthogonality formulation and Section 3 apply verbatim to semi-magic, pandiagonal and Franklin squares [12], with W replaced by the span of the relevant line indicators; Loly's inertia invariance already covers the semi-magic case. The hierarchy, energies and swap formulas for these families have not been worked out.

7 Conclusion

Reading the entries of a magic square as a height function on the centred grid makes the magic conditions the orthogonality of that function to a $2n$-dimensional space of test functions. We have treated this as a viewpoint, not a result, and used it to obtain a moment hierarchy whose degree-two level implies Loly's inertia invariance and whose permutation threshold drops below

the Vandermonde value $n-1$ from $n=14$ on; a balance theorem for the signed mass, with a sharp quantitative form and a negative result for unit-vector balance; exact moments and a limit law for the energies of a random arrangement, and the null distribution of Loly's moment of inertia; an exact single-swap formula with gap spectrum d^2m/n^4; and the exact count 14,639,440 of 4×4 arrangements satisfying the four low-mode conditions. What remains is a small set of exact statements about a classical object, together with the polyhedral companion paper A [14] and the open questions of Section 6.

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A Computational methods

Hardware and software. All computations were run on an Apple M4 Max (16 cores, 128 GB) under macOS, with Python 3.9.6, NumPy 2.0.2, SciPy 1.13.1 and C code compiled with Apple clang 17. Uniformly random arrangements were generated by Fisher–Yates shuffles (NumPy's default generator; `xoshiro256**` in C); the Monte Carlo of Table 3 uses 10^8 arrangements per order. All energies are evaluated from (5). The exact statements of Section 3 and the numbers quoted in Section 5 are checked in exact integer or rational arithmetic by `anc/verify_numbers.py`: the $n = 3$ enumeration, all 105,888 single swaps of the $880 + 8$ magic squares of orders 4 and 3 and the swaps of the Siamese squares, the exact means and variances for $n = 3, \dots, 6$, the null distribution of Loly's invariant, the witnesses, the product-of-inertia values, and the data for Section 5.4. The numerical statements of Section 4 are produced by `anc/balance_check.py`.

Exact moments and variances. The means in Proposition 5.1 are traces of the covariance matrix $\sigma_n^2 G$. The variances were computed as exact rationals for each $n \le 45$ by expanding $\mathbb{E}[E^2]$ over the 15 equality patterns of a cell quadruple, with the fourth-moment expectations of the uniform permutation expressed through the power sums p_2, p_4 of the centred values; the rational functions (22) were then obtained by interpolation separately for odd and even n and verified on 5–11 further orders each, and against the exhaustive $n = 3$ values. Densities of the Gaussian-limit mixtures in Figure 3 were computed by numerical inversion of the characteristic function, the method of Imhof [10].

Meet-in-the-middle enumeration of L_4. The four statistics $T_1 = \sum_{ij} j\, s_{ij}$, $T_2 = \sum_{ij} i\, s_{ij}$, $T_3 = \sum_i s_{ii}$, $T_4 = \sum_i s_{i,n-1-i}$ are additive over cells. The 16 cells are split into the top two rows A and the bottom two rows B. For each of the $\binom{16}{8} = 12{,}870$ value sets V and each of the 8! bijections $V \to A$ the partial vector $(T_1, \dots, T_4)|_A$ is packed into a 27-bit key; for the complementary set and each bijection $V^c \to B$ the key of (target − partial) is formed; the two key lists are sorted and merged, and the sum over matching keys of the product of multiplicities is $|L_4|$. This touches $2 \times 12{,}870 \times 40{,}320 \approx 1.04 \times 10^9$ partial vectors instead of $16! \approx 2.1 \times 10^{13}$ arrangements and runs in 9 s on 16 threads. Matching pairs are expanded explicitly to obtain the 14,639,440 members,

which are then classified by row and column sums and by D_4-orbit. Validation is described after Proposition 5.4.

Reproducibility. The scripts for every table and figure, the C enumerator and Monte Carlo code, the verification scripts and the JSON files containing the reported numbers are provided as ancillary files in `anc/`, with a `README` listing the commands that regenerate them.

B Variance closed forms

For a uniformly random arrangement the variances of the two energies of Proposition 5.1 are rational functions of n of the form

$$\begin{aligned} \operatorname{Var}(E_{\mathrm{low}}) &= \frac{(n-1)(n^2+1)\,P_{\mathrm{o}}(n)}{129600\,n^6}\ (n \text{ odd}), \qquad \frac{(n^2+1)\,P_{\mathrm{e}}(n)}{129600\,n^4}\ (n \text{ even}); \\ \operatorname{Var}(E_{\mathrm{full}}) &= \frac{(n-1)(n^2+1)\,Q_{\mathrm{o}}(n)}{180\,n^6}\ (n \text{ odd}), \qquad \frac{(n-2)(n^2+1)\,Q_{\mathrm{e}}(n)}{180\,n^5}\ (n \text{ even}), \end{aligned} \tag{22}$$

with integer polynomials P_{o}, P_{e} of degrees 11, 10 and leading coefficient 25, and Q_{o}, Q_{e} of degrees 6, 5 and leading coefficient 5, so that $\operatorname{Var}(E_{\mathrm{low}}) = n^8/5184 + O(n^7)$ and $\operatorname{Var}(E_{\mathrm{full}}) = n^3/36 + O(n^2)$. The polynomials are listed in the ancillary file `anc/data_witnesses_variances.json` (and in `anc/variance_fits.json`); they were obtained as described in Section A, and at $n = 3$ they give 1532/2187 and 398/2187, the exhaustive values, at $n = 4$ 14093/1440 and 527/640 (Table 3).

C A witness for $k^*(5) = 4$

The arrangement

$$\begin{pmatrix} 1 & 22 & 11 & 24 & 7 \\ 23 & 3 & 12 & 2 & 25 \\ 18 & 6 & 19 & 17 & 5 \\ 4 & 21 & 14 & 10 & 16 \\ 20 & 9 & 15 & 8 & 13 \end{pmatrix}$$

has all row sums equal to $M(5) = 65$ and column sums $66, 61, 71, 61, 66$, i.e. $\gamma = (1, -4, 6, -4, 1) = \left((-1)^j \binom{4}{j}\right)_j$; by (9) it satisfies the degree-3 conditions of Definition 3.10 without being semi-magic, so $k^*(5) = 4$. The analogous witnesses for $6 \le n \le 13$ (Remark 3.16) are in the ancillary file `anc/data_witnesses_variances.json`; `anc/verify_numbers.py` verifies them.